CS-663
Fundamentals of Digital Image Processing (DIP)

Introduction

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Digital Image Processing

- Digital image
  - N-dimensional array of numbers
- Algorithms
  - Input = image
  - e.g.,
    black-and-white picture (2D array of scalars),
    color picture (2D array of RGB vectors, 3D array of scalars)
    color video (multi-dimensional array of scalars)
  - Output = image, or measurement, or high-level description
- Applications
  - Images are everywhere

Digital Image Processing

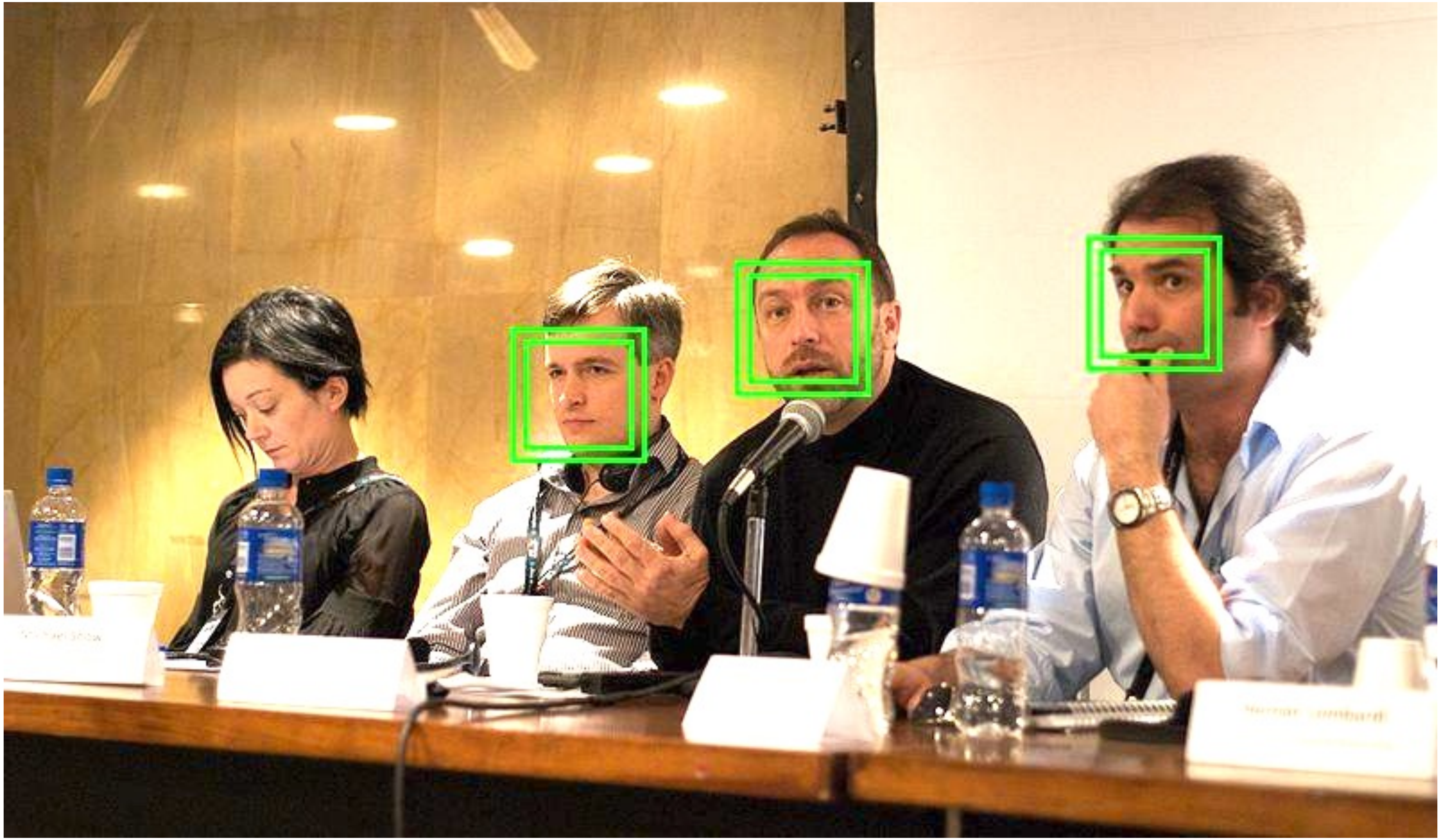
- Application domains
- Pictures, videos: cameras, smartphones, webcam
- Photography, cinema (editing, special effects)
- Medical image analysis: microscopy, X-ray, CT, MRI, endoscopy
- Remote sensing (weather prediction)
- Biometrics, forensics
- Surveillance (military, urban)
- Seismology
- Sports
- …

Digital Image Processing

- Why take this course ?
- Gentle introduction to key ideas in computational data analysis
- Gentle introduction to key ideas in computational image analysis
- Key ideas founded within:
  statistics, linear algebra, machine learning, deep neural networks
- Jobs in R&D
  - India and abroad
  - Adobe, Canon, Disney, ExxonMobil, GE, Google, Facebook, Hitachi, Hewlett Packard, Medtronic, Microsoft, Nvidia, Intel, Philips, Qualcomm, Samsung, Siemens, Sun Pharma, Toshiba, Waymo
  - Lots of startups

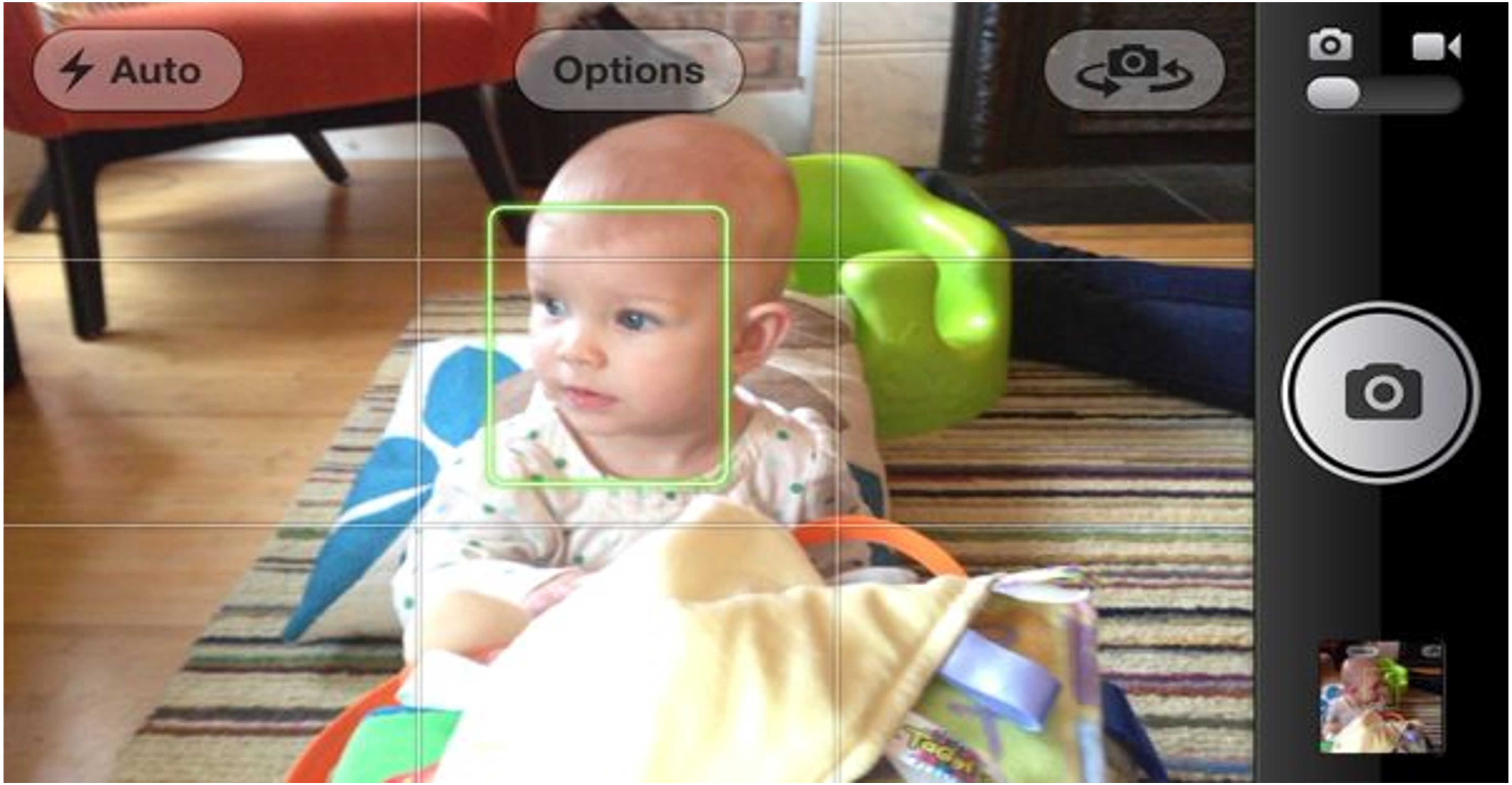
Digital Image Processing

- en.wikipedia.org/wiki/Face_detection



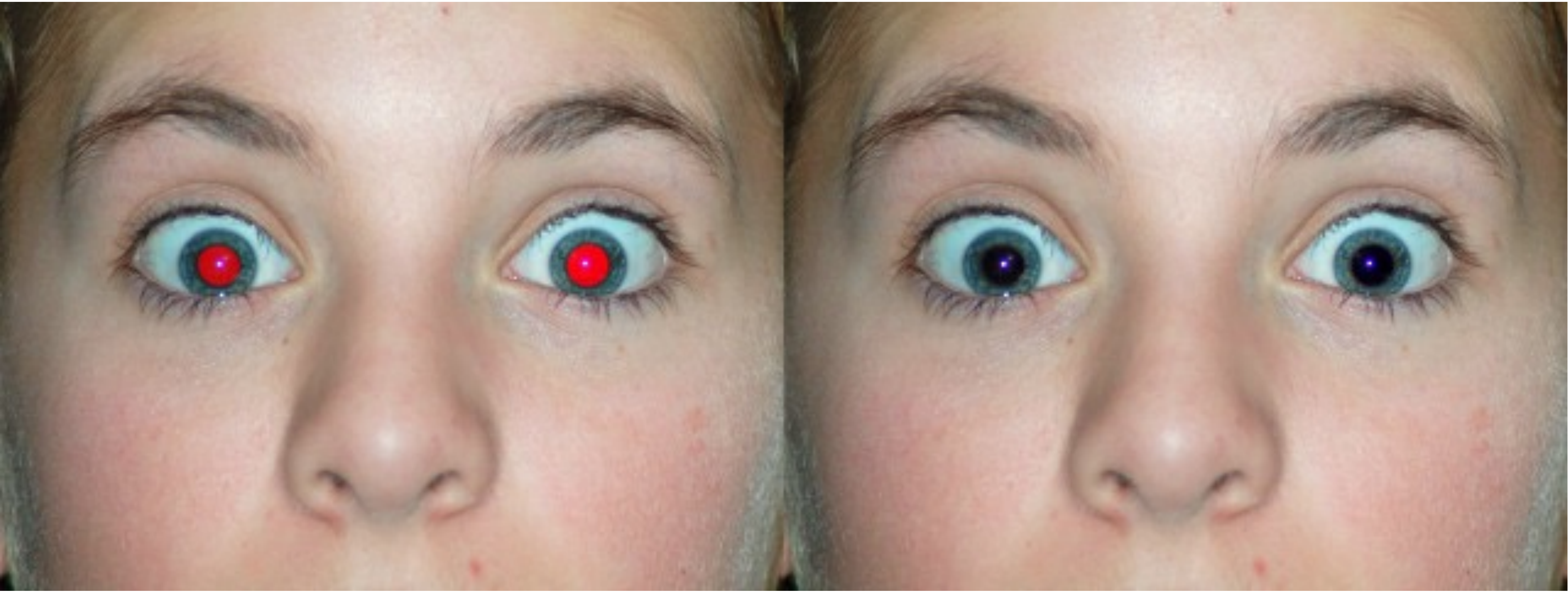
Digital Image Processing

- en.wikipedia.org/wiki/Autofocus



Digital Image Processing

- Red-eye correction
- en.wikipedia.org/wiki/Red-eye_effect



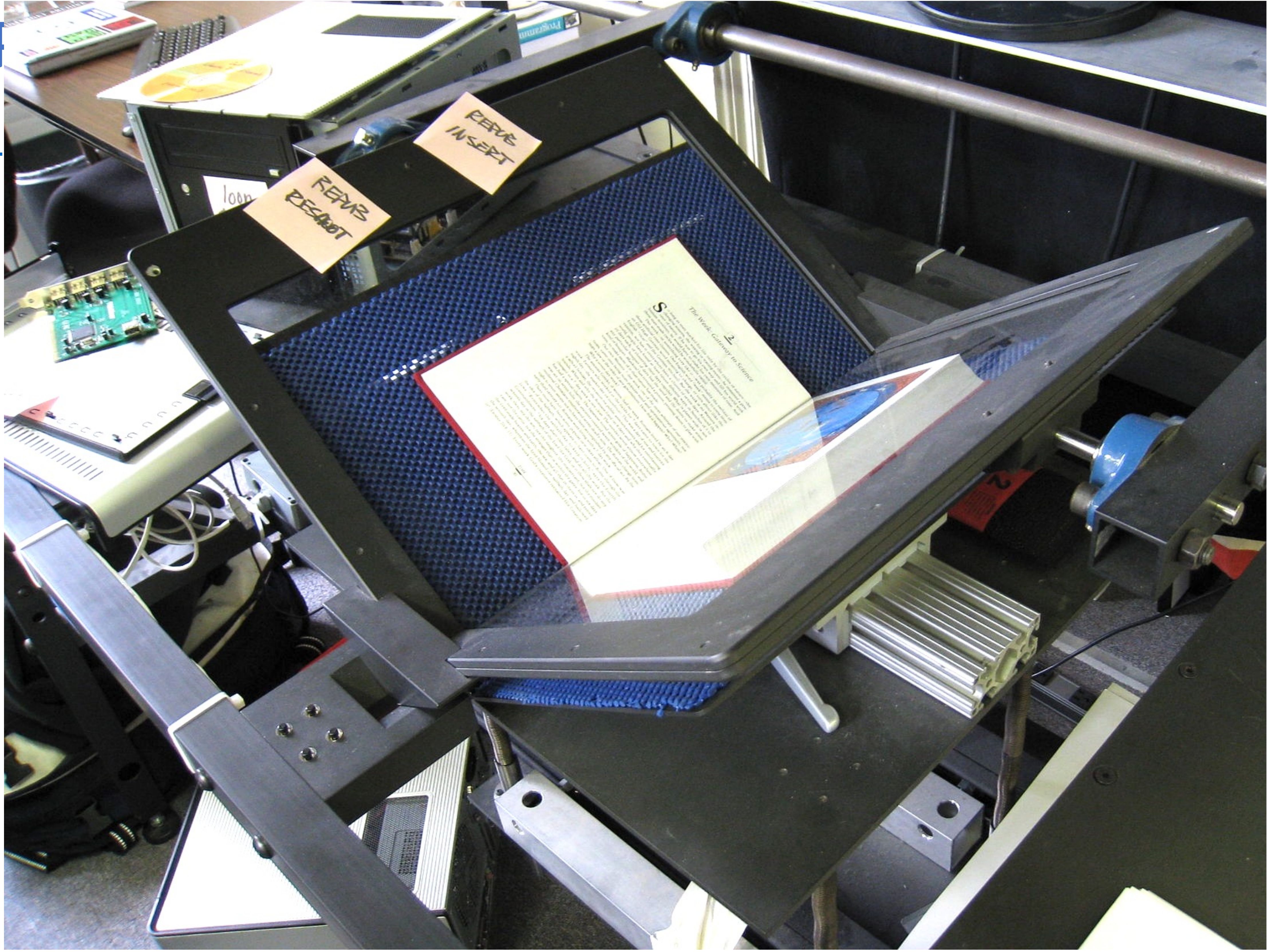
Digital Image Processing

- en.wikipedia.org/wiki/Panoramic_photography

- en.wikipedia.org/wiki/Panoramic_radiograph

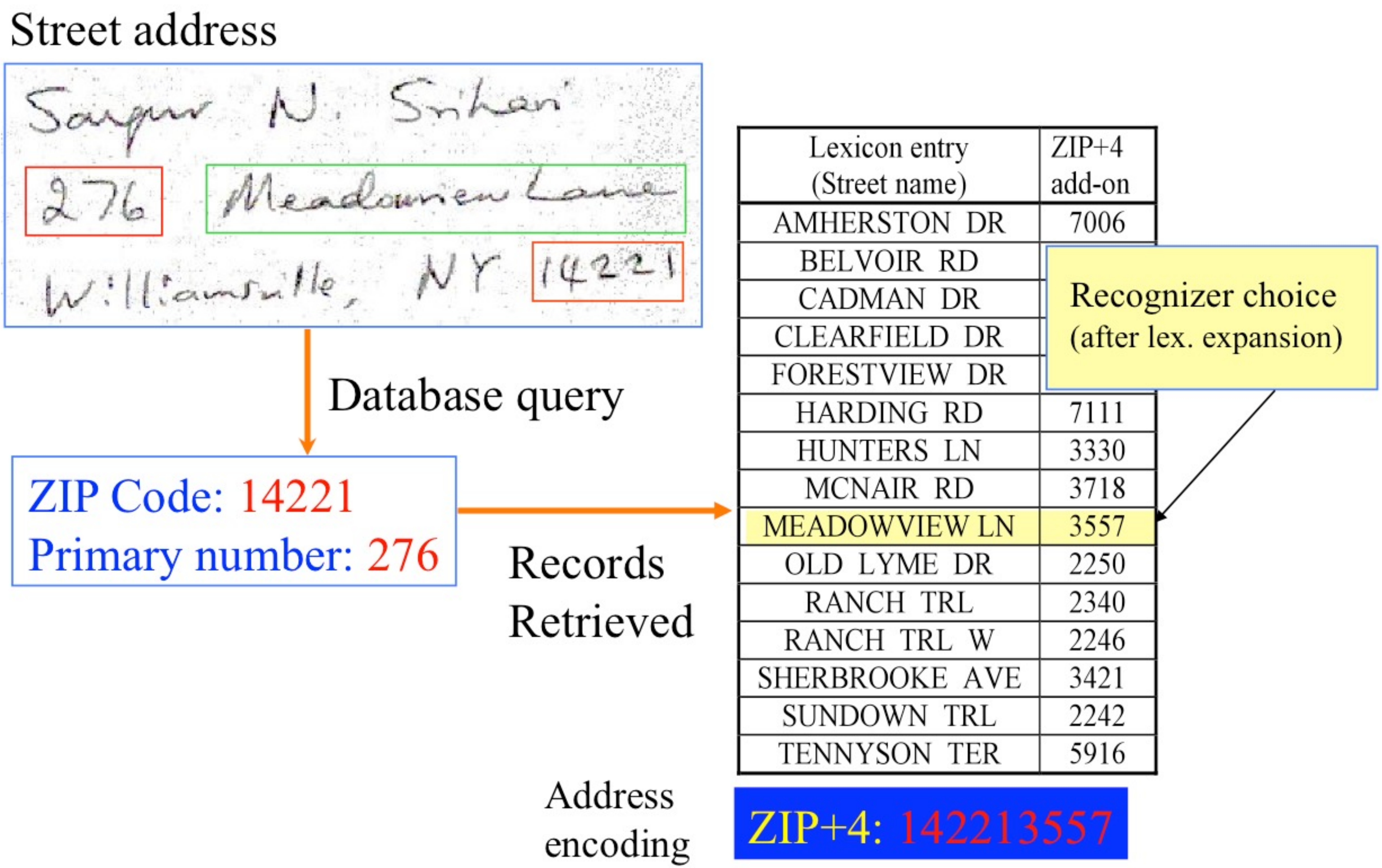
Digital Image

- en.wikipedia.org/wiki/Digitization



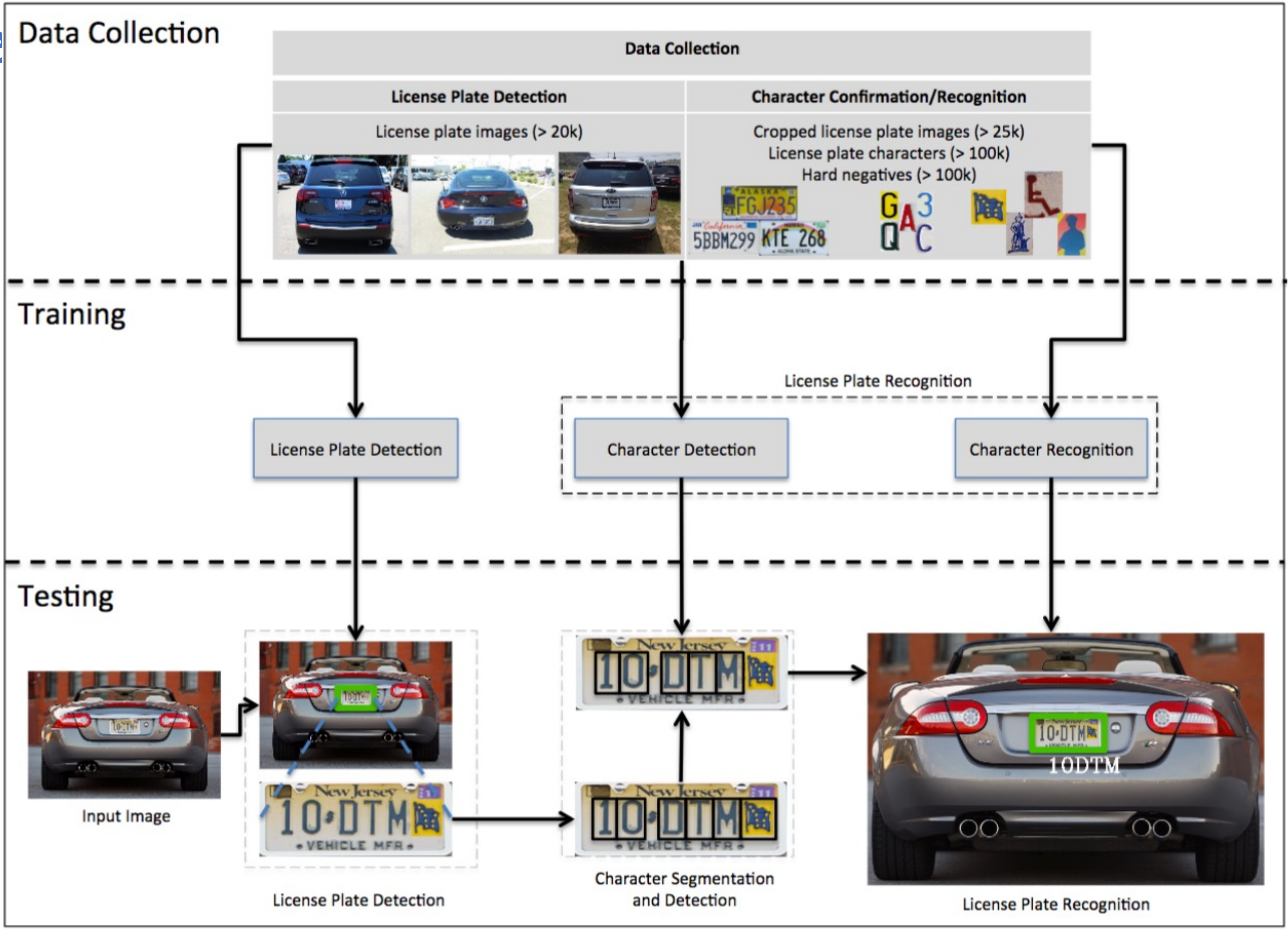
Digital Image Processing

- en.wikipedia.org/wiki/Handwriting_recognition



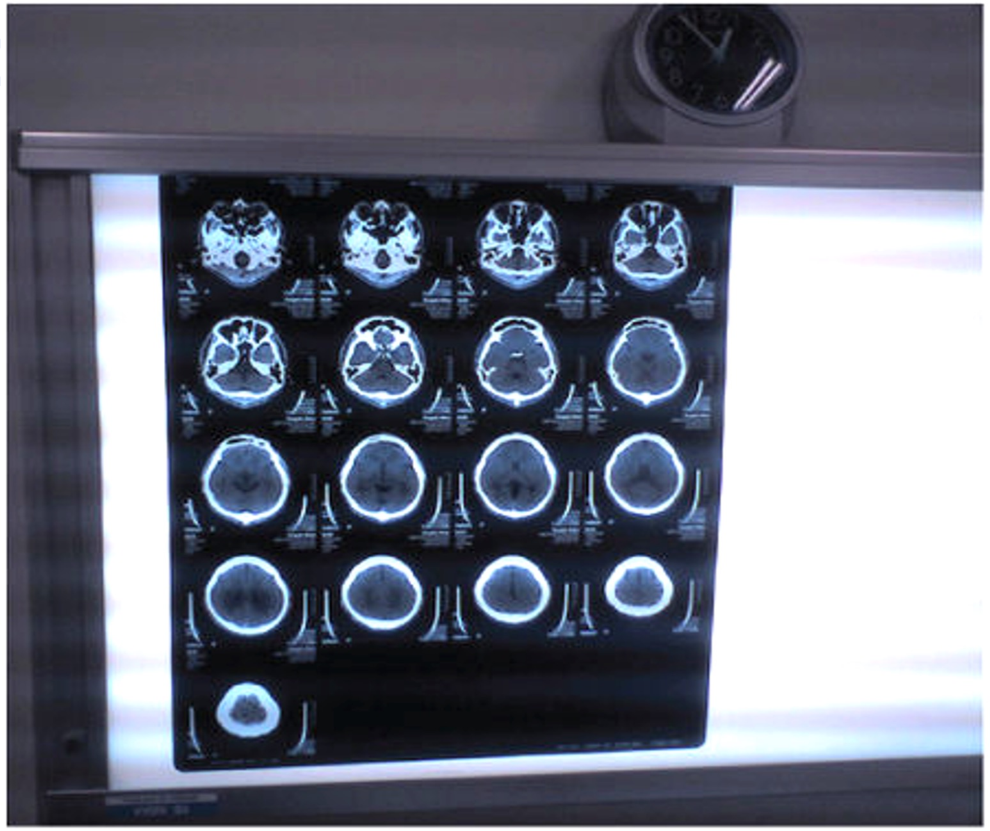
Digital Image

- [en.wikipedia.org/wiki/Automatic number-plate recognition](https://en.wikipedia.org/wiki/Automatic_number-plate_recognition)



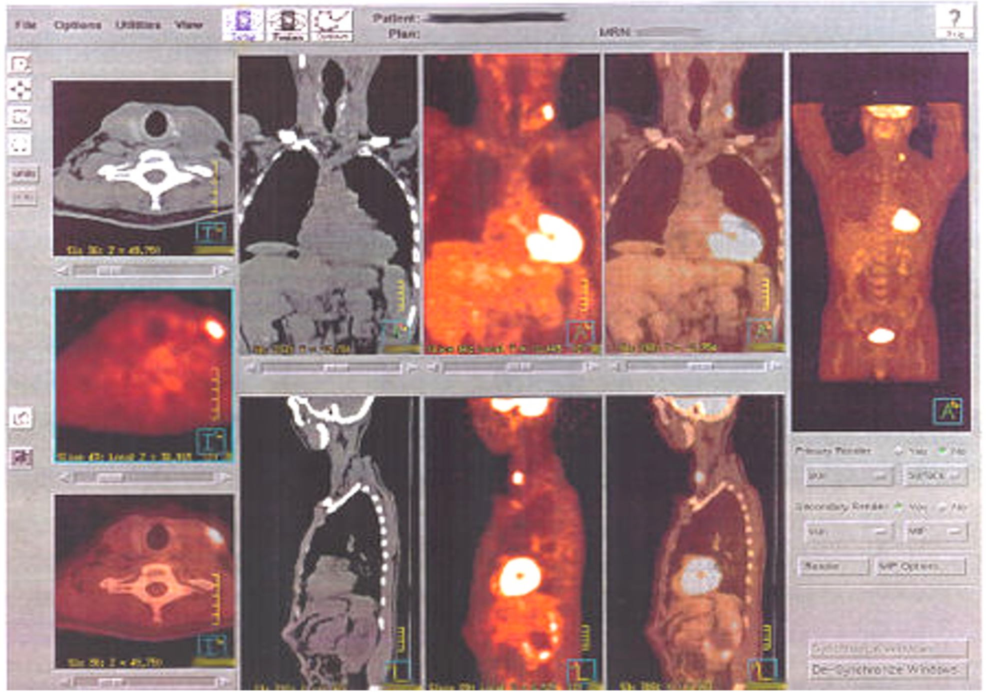
Digital Image F

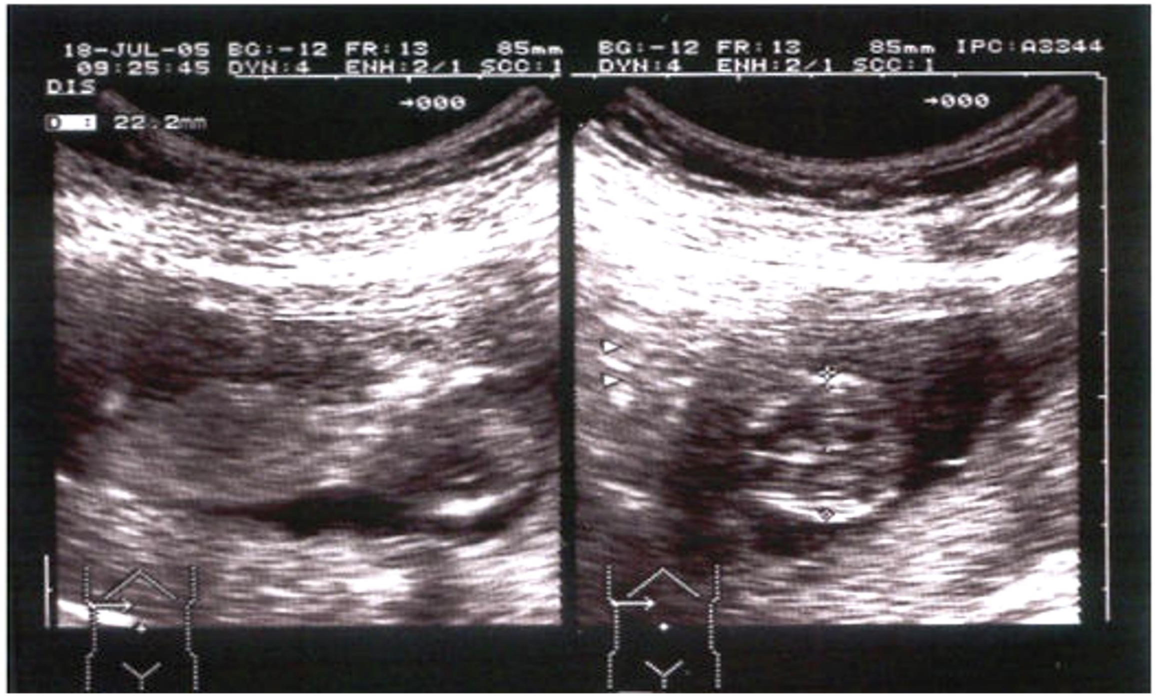
- [Medical imaging](): Xray, CT, PET, MRI, Ultrasound, Microscopy, Laparoscopy
  - Scanning process in modalities like CT, PET, MRI is digital by design



(a)

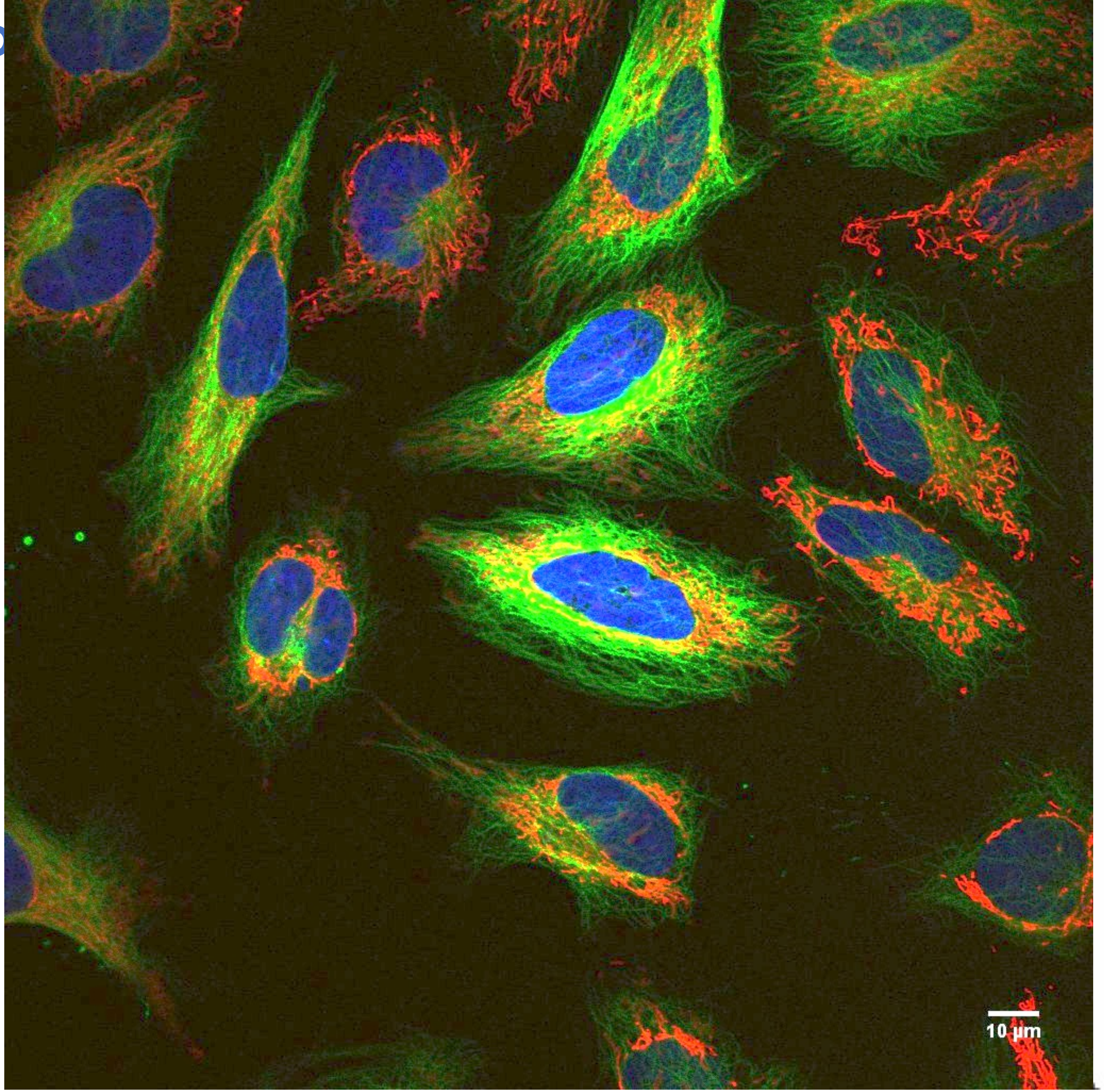
(b)





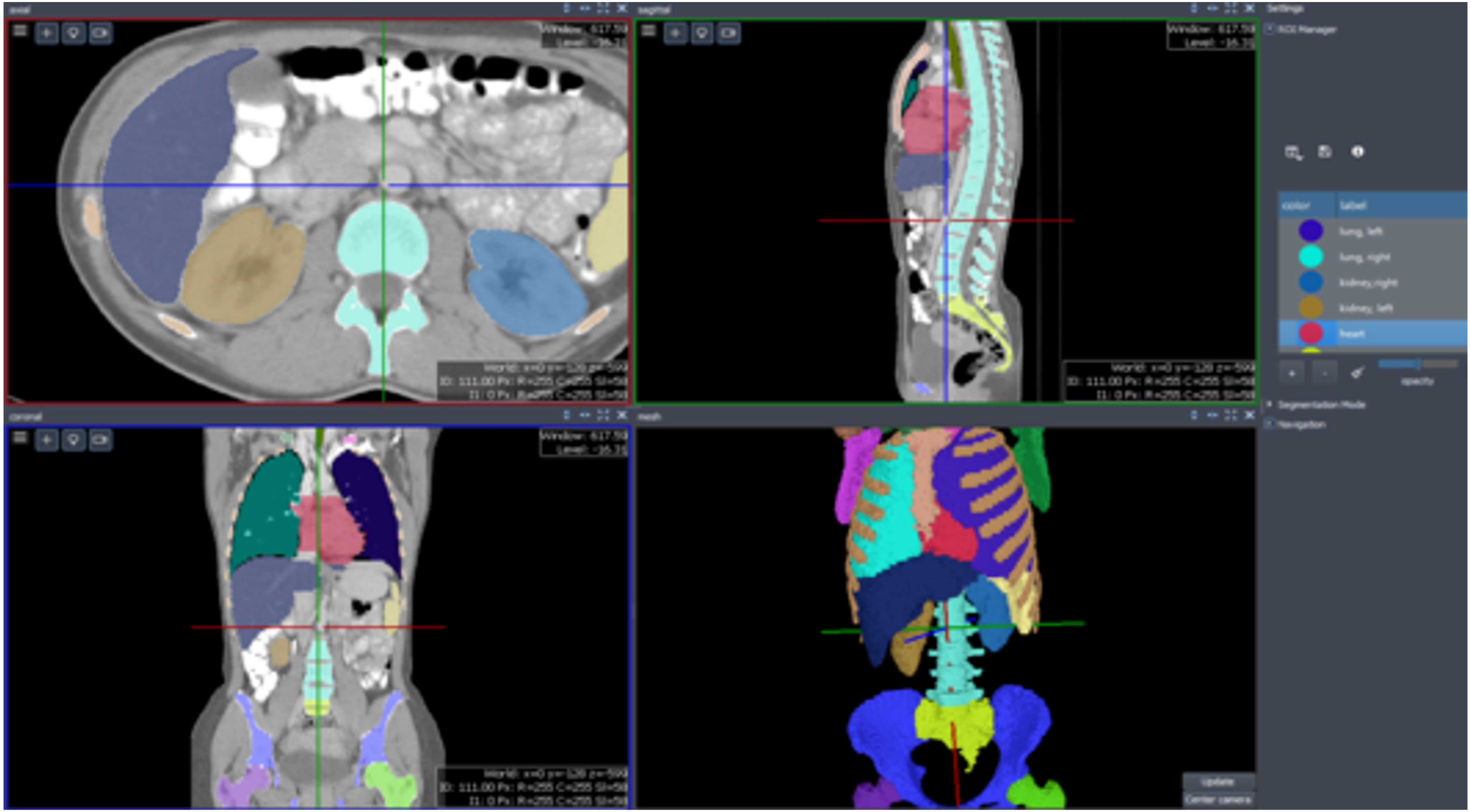
Digital Image P

- [Biological imaging]
- [Preclinical imaging]



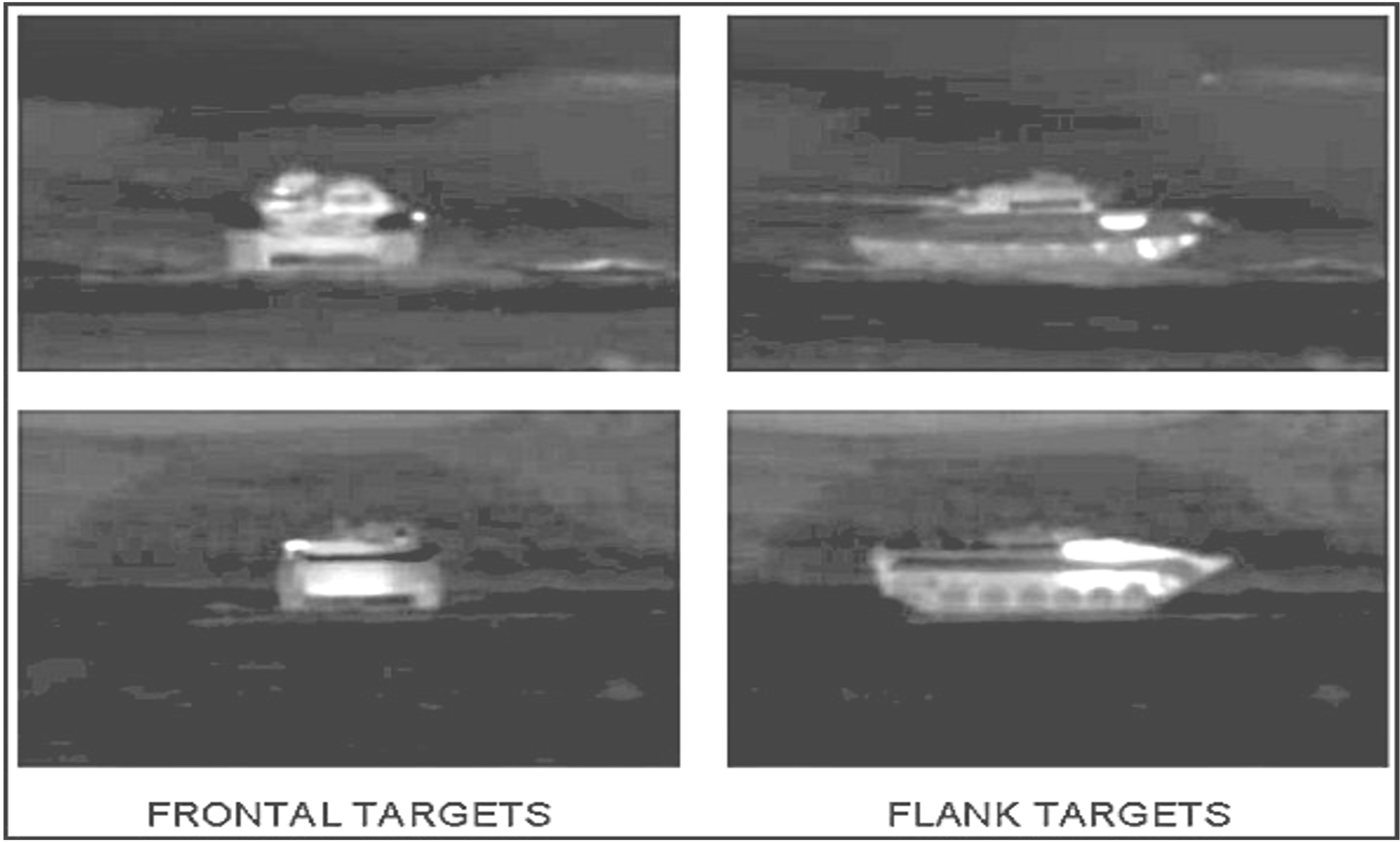
Digital Image Processing

- en.wikipedia.org/wiki/Medical_image_computing



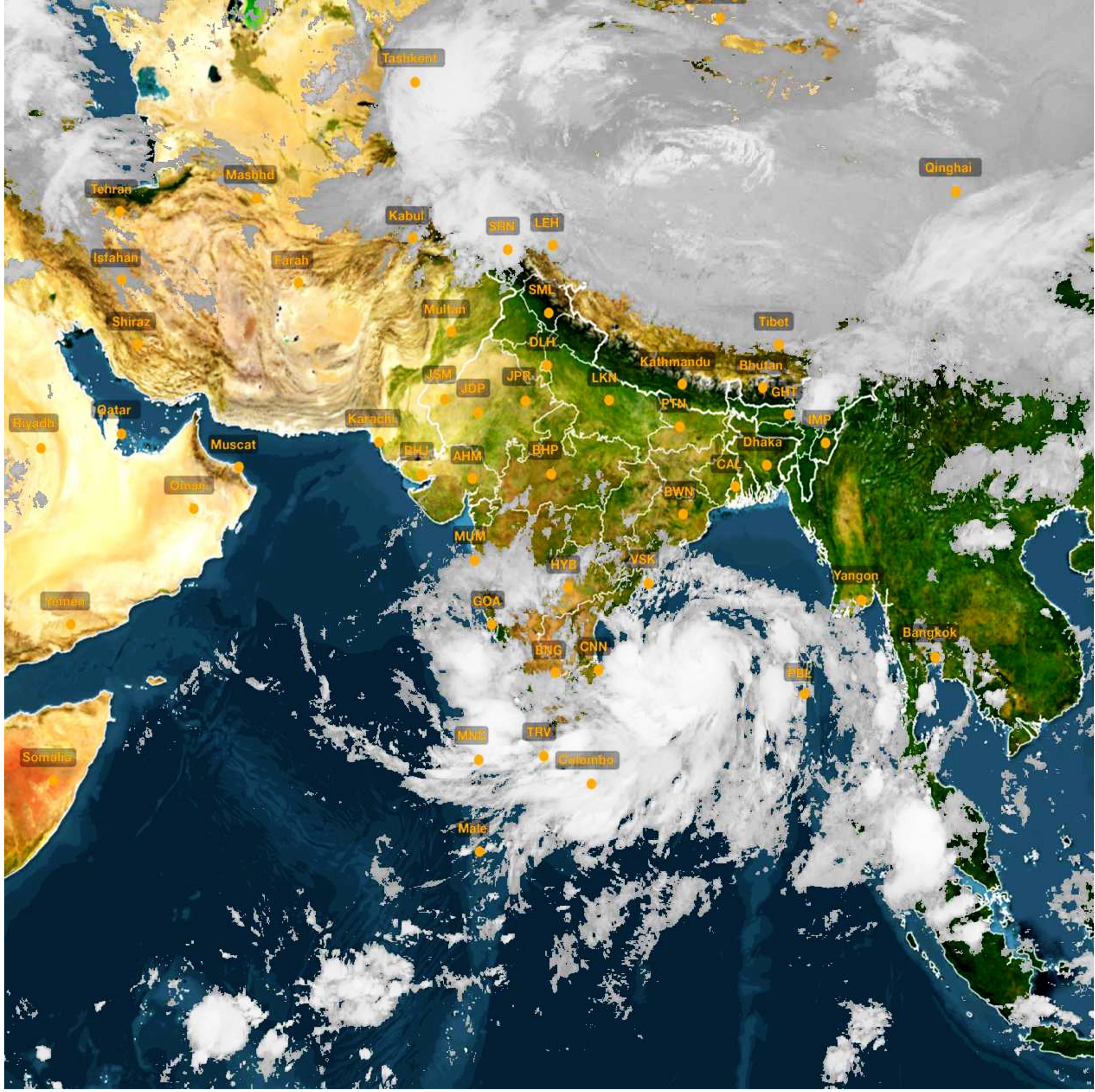
Digital Image Processing

- Defence



Digital Image Processing

- en.wikipedia.org/wiki/Remote_sensing
- Prediction weather
- Track, predict movement of cyclone



Digital Image Pr

- Plant identification
- Based on image of leaf, flower, fruit, stem, or entire plant
- identify.plantnet.org

Digital Image Processing

- Agriculture : Classification of plant diseases


Apple scab


Cherry powdery mildew


Corn northern leaf blight


Grape black rot


Grape leaf blight


Orange Haunglongbing (Citrus greening)


Peach bacterial spot


Potato early blight


Squash powdery mildew

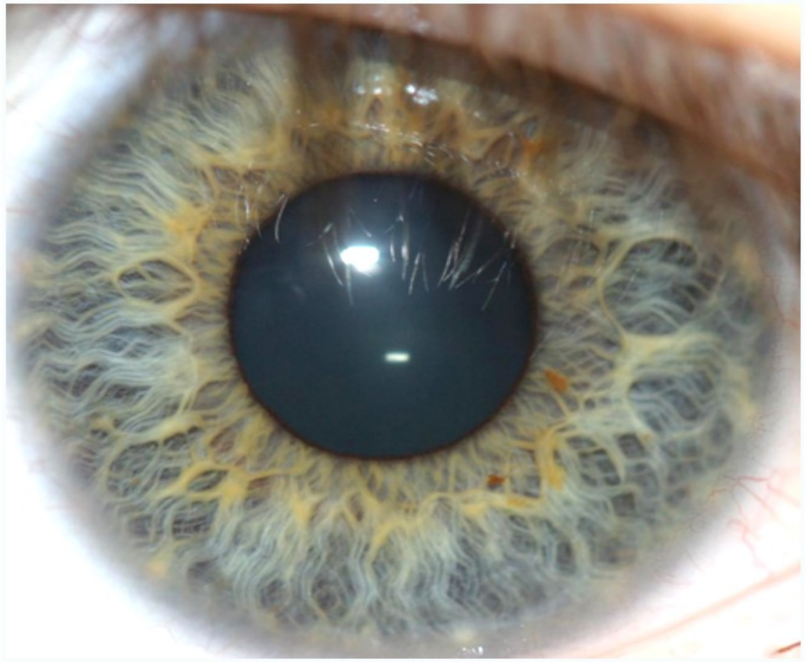
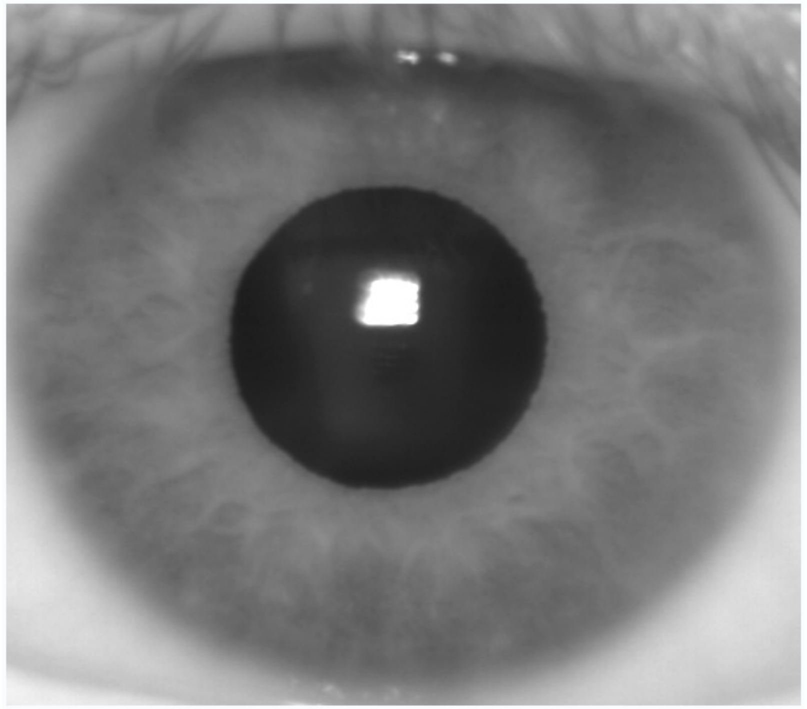
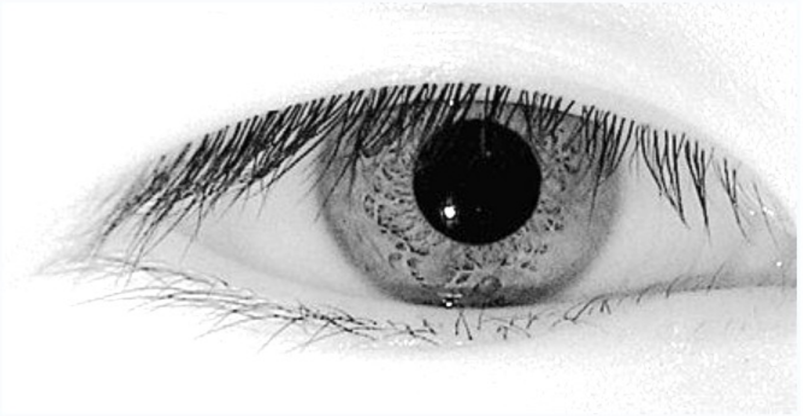

Strawberry leaf scorch


Tomato early blight


Tomato late blight

Digital Image Processing

- Biometrics
- en.wikipedia.org/wiki/Iris_recognition
- aadhaar.bioenabletech.com/aadhaar-iris-scanner

Visible wavelength iris image	Near infrared (NIR) version	NIR imaging extracts structure
 Visible light reveals rich pigmentation details of an Iris by exciting melanin, the main colouring component in the iris.	 Pigmentation of the iris is invisible at longer wavelengths in the NIR spectrum	 Even "dark brown" eyes reveal rich iris texture in the NIR band, and most corneal specular reflections can be blocked.

Digital Image Processing

- en.wikipedia.org/wiki/Crowd_counting
- www.indiatoday.in/india/story/artifical-intelligence-demystifying-crowd-counting-process-prayagraj-2685514-2025-02-25

Digital Image Processing

- en.wikipedia.org/wiki/Image_editing
  - Generating texture

Digital Image Processing

- en.wikipedia.org/wiki/Image_editing
    - Inpainting

Digital Image Processing

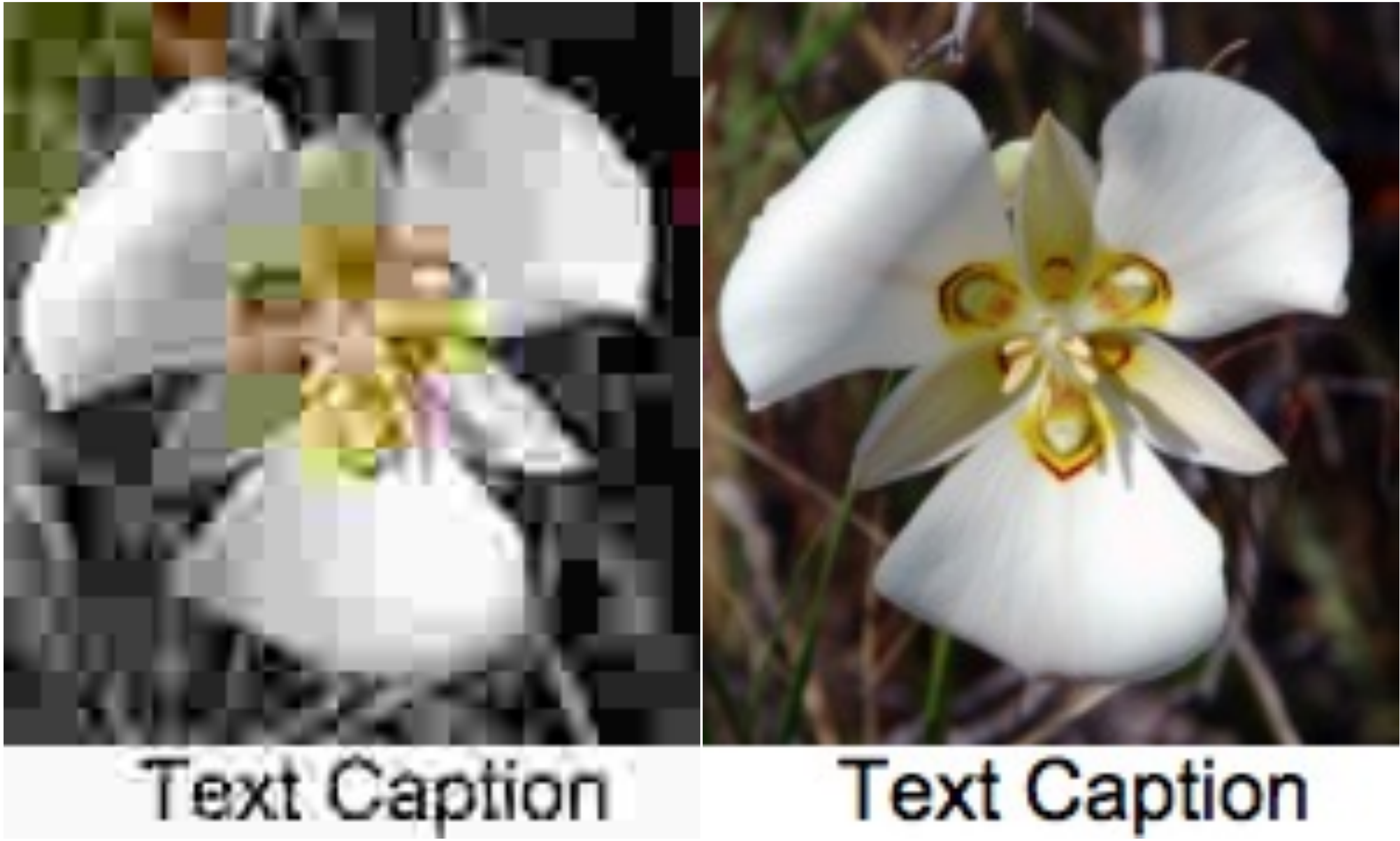
- en.wikipedia.org/wiki/Digital_photograph_restoration

Digital Image Processing

- en.wikipedia.org/wiki/Digital_photograph_restoration

Digital Image Processing

- Image and video compression



Digital Image Processing: Evaluation Structure

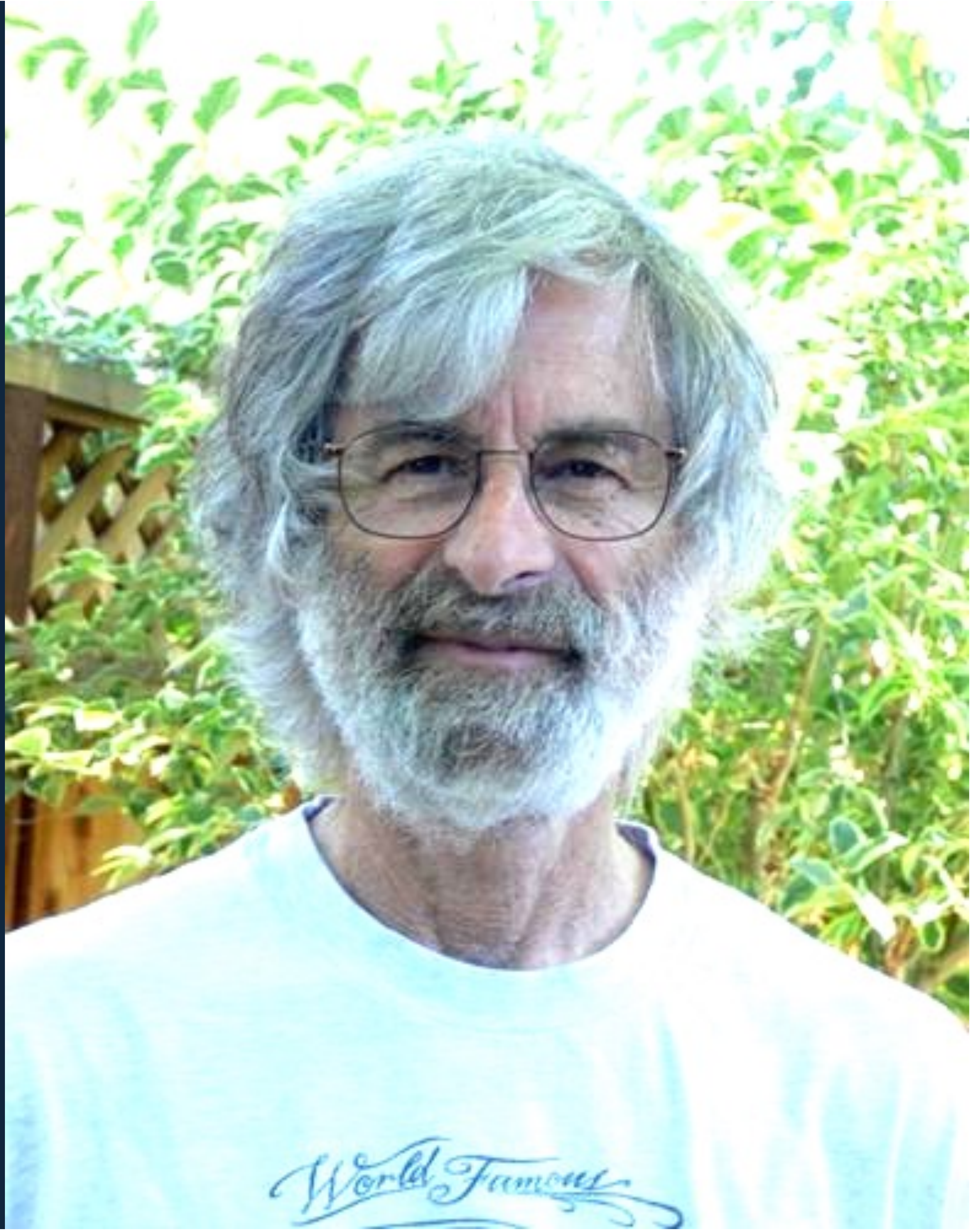
- Assignments: 3 (around 25%)
  - Programming based
- Quiz ?
- Exams: 2 (around 55%)
  - Mid-semester exam
  - End-semester exam
- Course project: 1 (around 15%)
- Class participation: around 5%
- "around" = ±10% is allowed, e.g., 50% → 40%, 15% → 25%

- "Audit" registration: Not Allowed

Digital Image Processing

"If you're thinking without writing, you only think you're thinking. [...] Everyone thinks they think. If you don't write down your thoughts, you are fooling yourself."

— Leslie Lamport, creator of LaTeX (a document prepration system)



Digital Image Processing

- Assignments
  - Implement algorithms in carefully curated (not wild) settings to learn concepts
    - Use any coding environment: Python, Matlab, OpenCV, Insight Toolkit (ITK)
    - Experiment with real-world and simulated image data
  - For maximum benefit, assignments must be done alone
    - Ask TAs, or instructor, about technical queries
  - If done in a group, each student in the group must solve each and every question, independently or jointly
    - e.g., cannot split work as:
      student A solves questions 1 and 2
      student B solves questions 3 and 4
    - This is important for effective learning

Digital Image P

- Plagiarism

Digital Image Processing

- Plagiarism policy
  - Institute policy www.iitb.ac.in/newacadhome/punishments201521July.pdf
    - “A student found copying in an assignment/laboratory project is given a zero in the assignment/project and is further given a one grade penalty.”
    - “The same disciplinary action is taken against both the person copying and the person from whom the material was copied.”
- CSE policy
  - Teaching assistant or instructor will report case to Departmental Academic Disciplinary Action Committee (DADAC)
  - At least loss of 1 grade level
    - Can get a fail grade

Digital Image Processing

- Plagiarism policy
  - https://integrity.mit.edu/handbook/writing-code
  - "…, during the time that you are helping another student, your own solution should not be visible, either to you or to them. Make a habit of closing your laptop while you're helping."
  - "As they type lines of code, they speak the code aloud to the other person, to make sure both people have the right code. INAPPROPRIATE."
  - "In a tricky part of the problem set, Alyssa and Ben look at each other's screens and compare them so that they can get their code right. INAPPROPRIATE."

Digital Image Processing

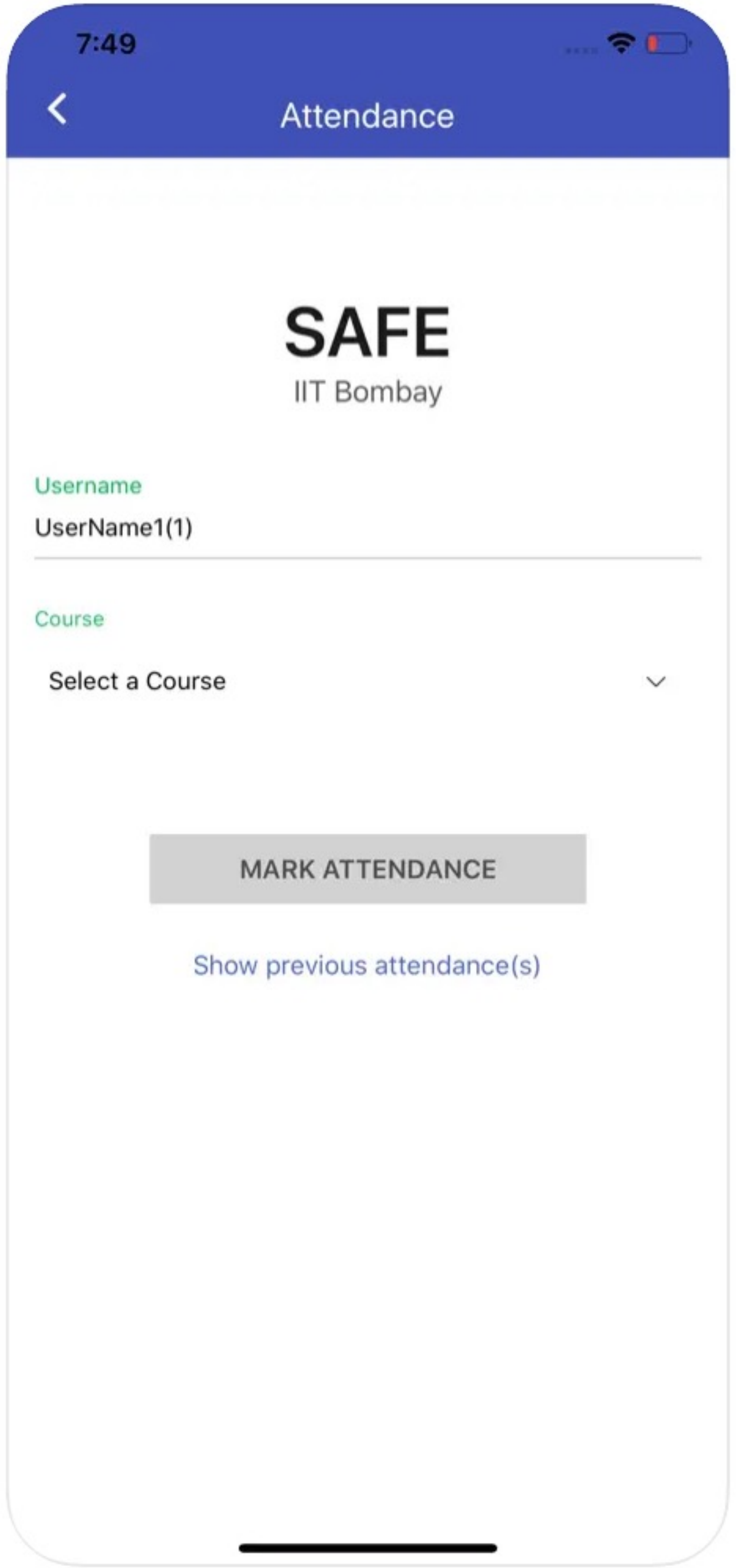
- Plagiarism policy https://integrity.mit.edu/handbook/writing-code
  - "Jerry opens his own laptop, finds his solution to the problem set, and refers to it while he's helping Ben correct his code.
    INAPPROPRIATE."
  - "Ben has by now spent a couple hours with Louis, and Louis still needs help, but Ben really needs to get back to his own work. He puts his code in a Dropbox and shares it with Louis, after Louis promises only to look at it when he really has to.
    INAPPROPRIATE."
  - "They exchange their test cases with each other to check their work.
    INAPPROPRIATE.
    Test cases are part of the material for the problem set, and part of the learning experience of the course. You are copying if you use somebody else's test cases, even if temporarily."

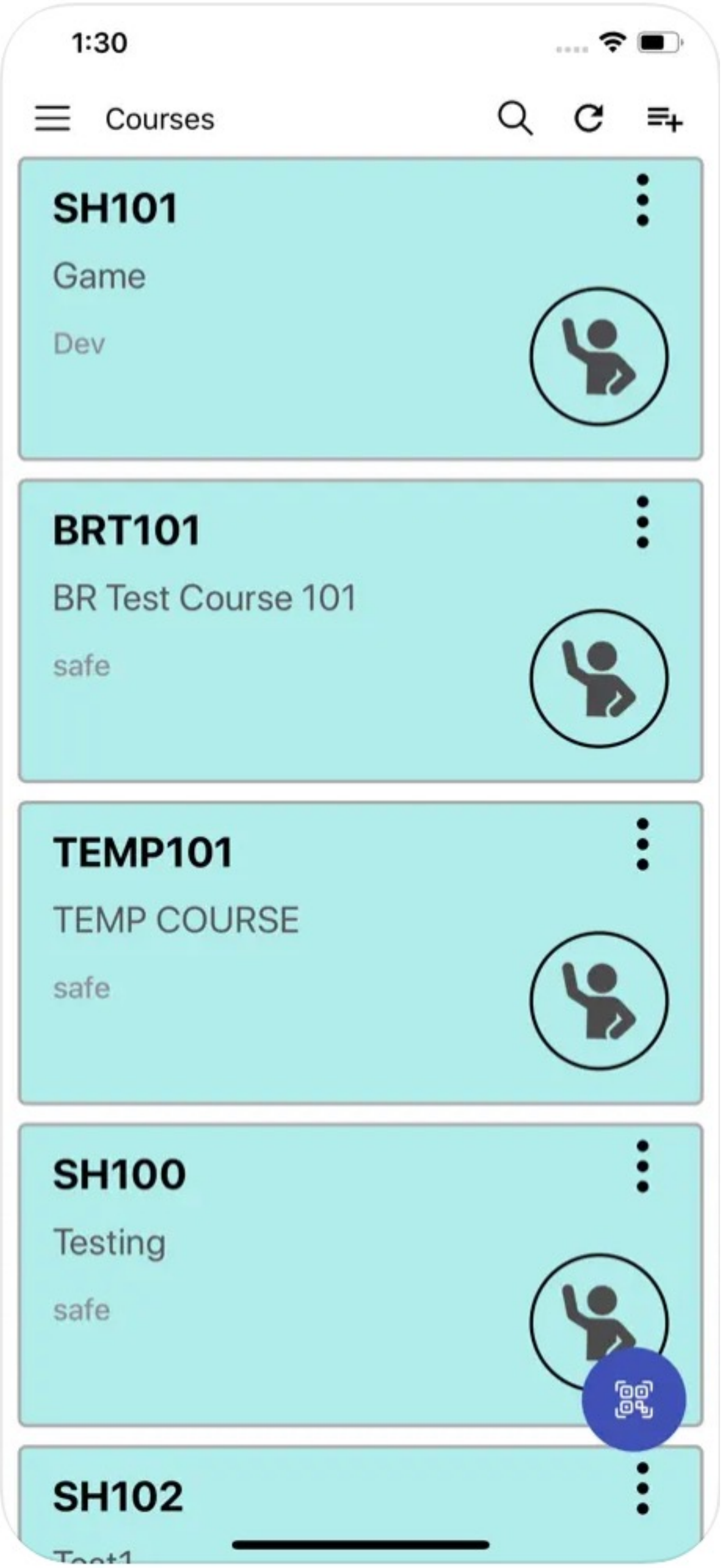
Digital Image Processing

- Attendance policy
  - https://www.iitb.ac.in/newacadhome/ugrulebook.pdf
  - https://www.iitb.ac.in/newacadhome/MTechRules.pdf
  - “Attendance in classes is mandatory from the very beginning of the semester. Students who miss even a single lecture from among the first three lectures of a course, are liable to have themselves deregistered from the corresponding course.”
  - "IIT Bombay expects one hundred percent attendance (100%) from its students in all classes. If the attendance of the student, as counted with effect from the first contact hour, falls below eighty percent of the total attendance expected, the instructor may award the student a ‘Drop due to inadequate attendance’ ‘DX’ grade in that course.”
    - “Only exception to this rule are courses where the instructor has declared that no DX grade will be awarded."
  - Medical certificate needed from IITB hospital, or one authenticated by them.

Digital Image Processing

- Attendance via SAFE app
- safe.cse.iitb.ac.in
- “How to Signup on SAFE App”: www.aero.iitb.ac.in/home/pdf/SAFE_Login.pdf

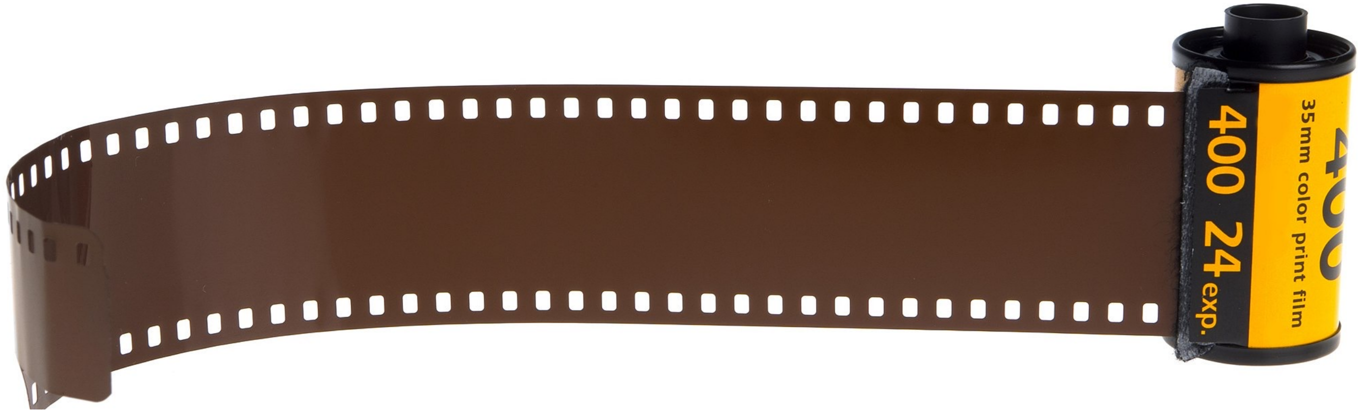




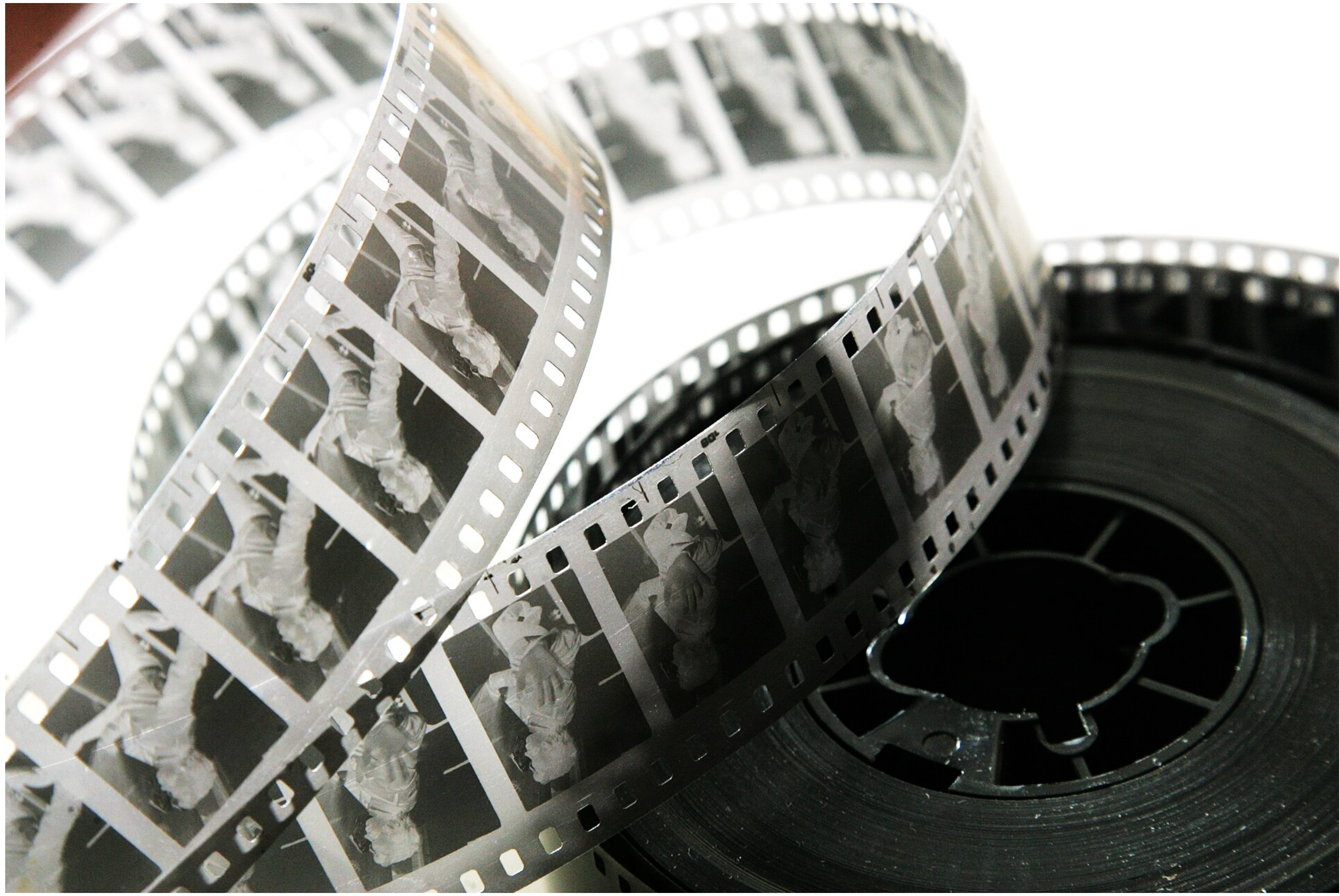
Digital Image Processing

- Signals can be modeled as functions
- Types of domains
  - Discrete-valued, continuous-valued
  - Time: temperature at a location in Mumbai, stock-market index
  - Space: image
  - Space + Time: video (no sound)
- Types of ranges
  - Discrete-valued, continuous-valued
  - Scalar-valued, vector-valued
- $1 \rightarrow 1$: stock-market index
- $2 \rightarrow 1$: black-and-white photograph
- $3 \rightarrow 3$: video (color RGB; no sound)

Digital Image Processing



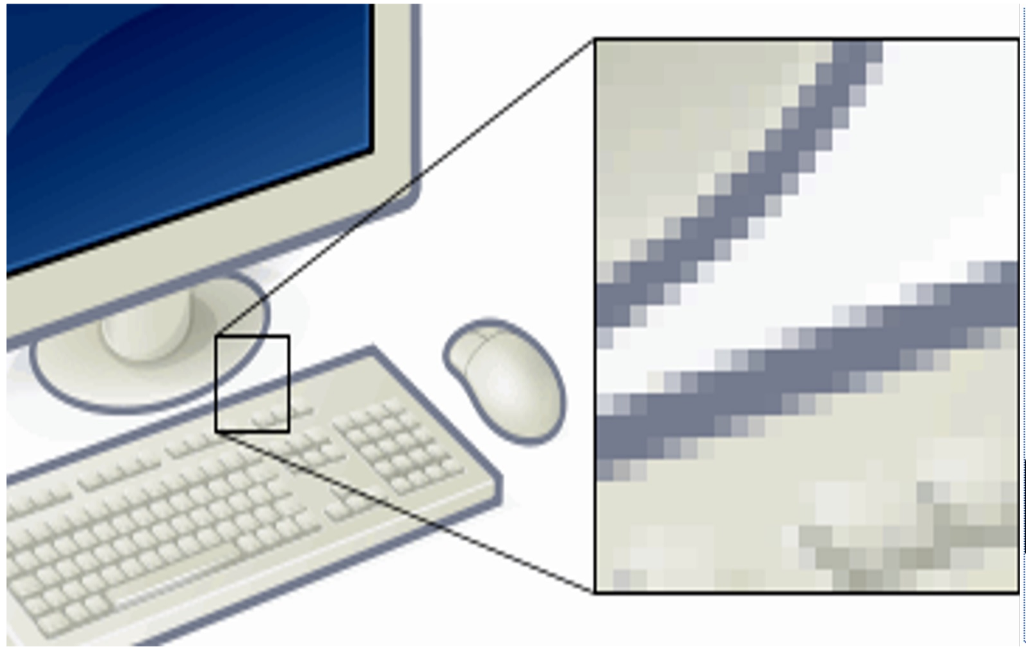
- “Analog” Image: Continuous function
- No discretization, infinite “resolution”
- Data acquired on [photographic film]() (“negative”)

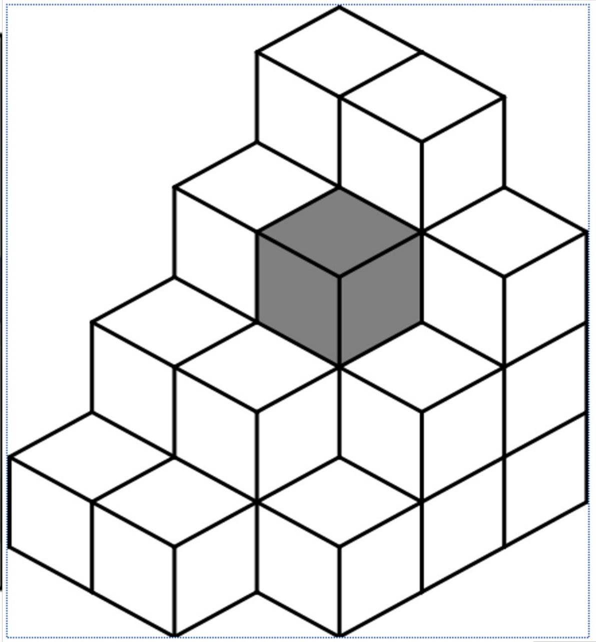


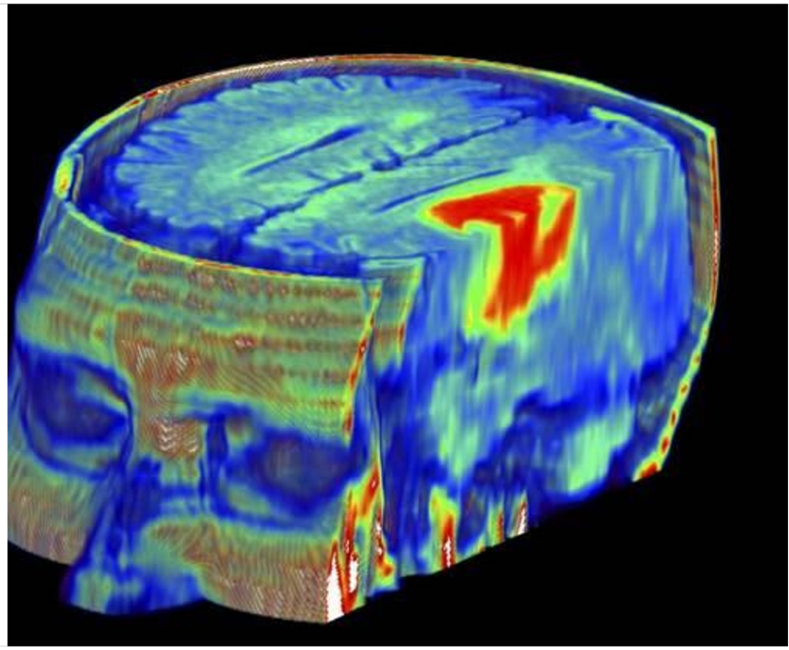
- This course
  isn't about analog images/signals

Digital Image Processing

- Digital (discrete) Image: array / grid of numbers
  - Can have one or more dimension
  - Some dimensions have semantics of physical space
- What makes data discrete ?
  - Acquisition: e.g., charge-coupled device (CCD) array in camera
  - Representation: e.g., digitization : analog print → scanned digital copy
- This course is about digital images
- Array element
  - Pixel: 2D
  - Voxel: 3D







Digital Image Processing

- Imaging = Process of acquiring/measuring image
  - Taking a digital photograph
  - Taking an X-ray / CT / MRI scan

1) Imaging Hardware / Instrumentation
   - Camera lens system, SLR, ...
   - X-ray machine, MRI machine, ...

2) Imaging Physics
   - Camera → Optics
   - Ultrasound, Seismic → Acoustics
   - MRI → Electromagnetism

Digital Image Processing

- This course isn't about imaging
- This course is about image processing
- Image data is given: scanner is a black box
- Don't know hardware details
  - e.g., what is inside photographic camera
- Don't know physics details
  - e.g., optics
- We know a high-level system specification
- We know transformation from physical quantity → data
  - e.g., objects and colors in scene → acquired image

Digital Image Processing

- Coding environments

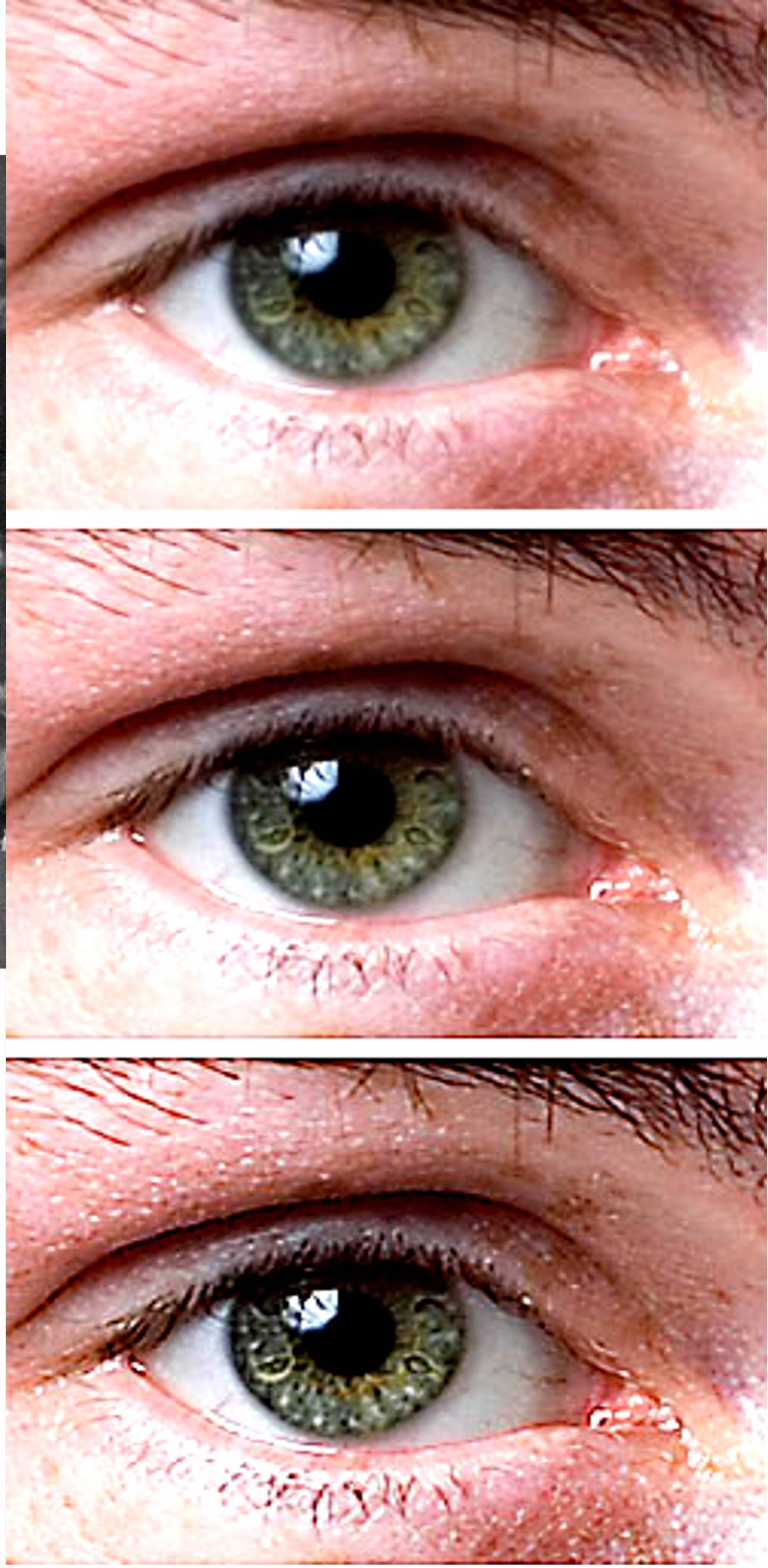
1) Python
    - Pytorch is a powerful framework for deep learning using neural networks
2) Matlab
    - Also supports deep neural networks
3) OpenCV, Insight Toolkit
    - Based on C++

Digital Image Processing

- Matlab Tutorial 1
  - www.mccormick.northwestern.edu/documents/students/undergraduate/introduction-to-matlab.pdf
- Matlab Tutorial 2
  - http://web.eecs.umich.edu/~aey/eecs451/matlab.pdf
- Matlab Tutorial 3
  - www.cs.cmu.edu/~ggordon/10601/recitations/matlab/pretty_matlab_pres.pdf
- Matlab Image-Processing Toolbox Tutorial
  - http://www.cs.otago.ac.nz/cosc451/Resources/matlab_ipt_tutorial.pdf
- Writing fast matlab code
  - http://www.csc.kth.se/utbildning/kth/kurser/DN2255/ndiff13/matopt.pdf
- Matlab Tips and Tricks
  - http://www.ee.columbia.edu/~marios/matlab/mtt.pdf

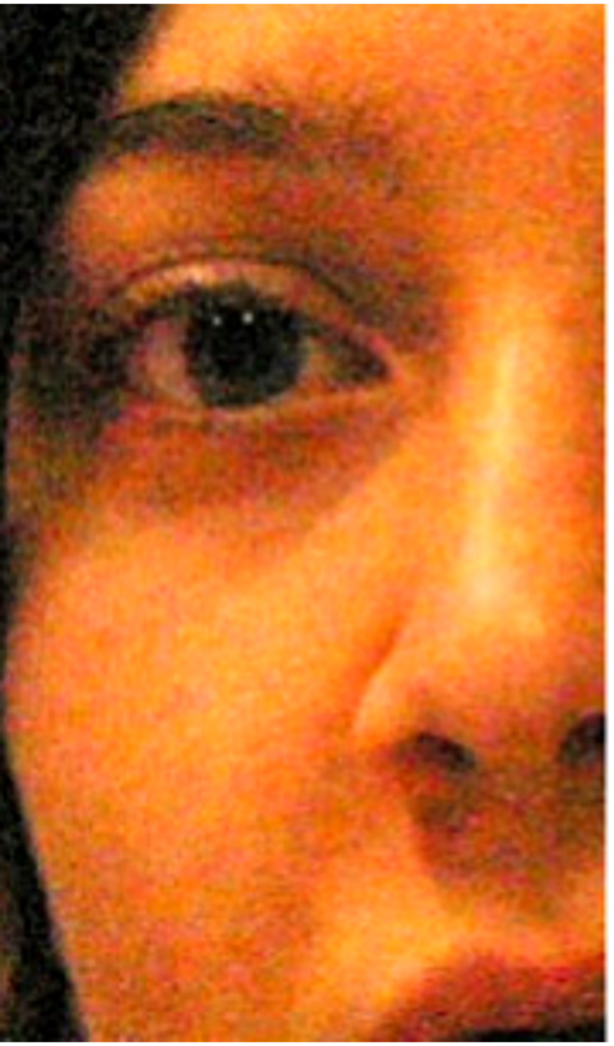
Digital Image Processing: Applications

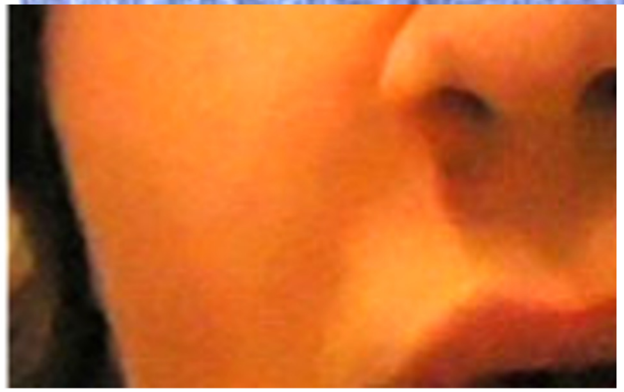
- Image enhancement
- Improve contrast
- Sharpen



Digital Image

- Image denoising



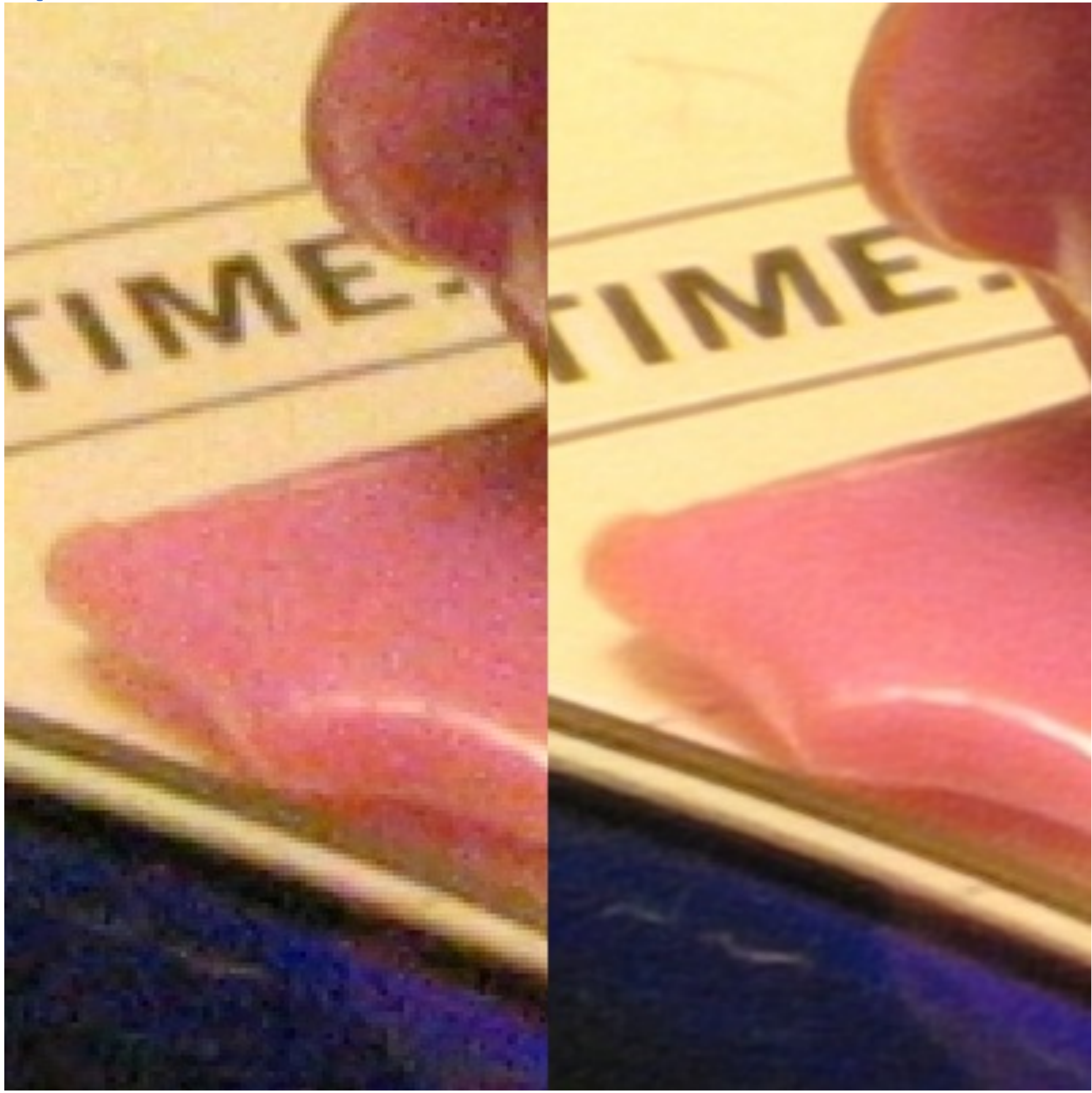


Digital Image Processing: Applications

- [Image restoration]()

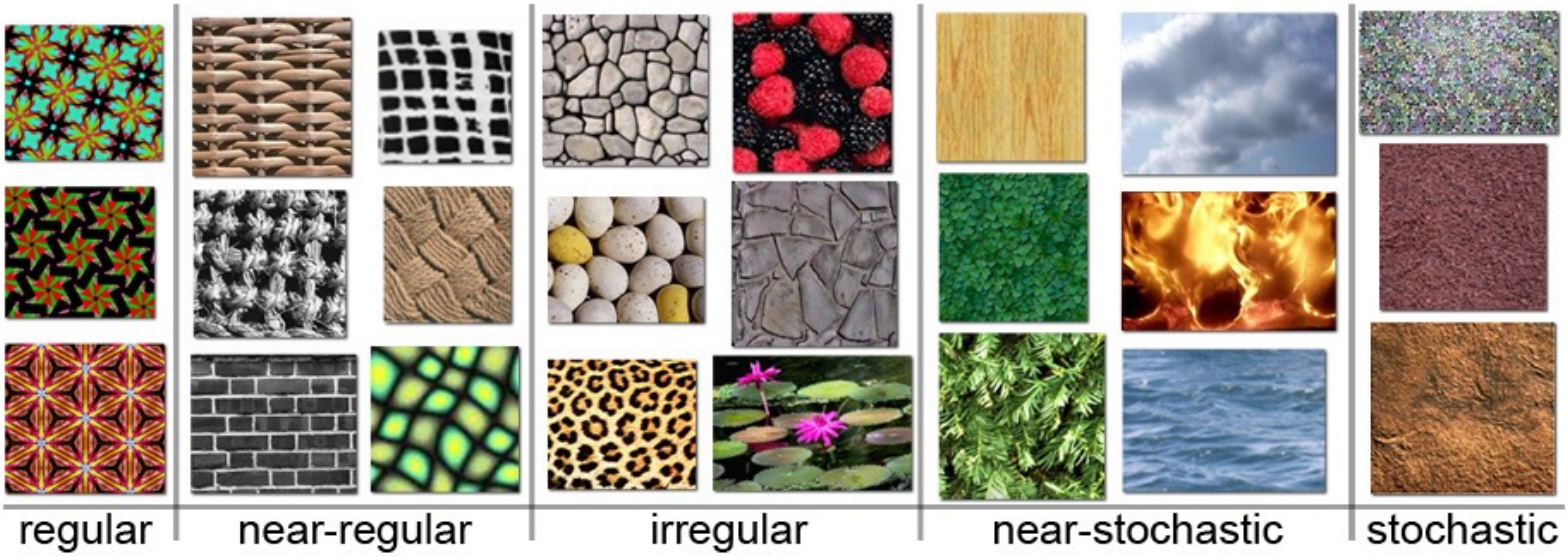
Digital Image Processing: Applications

- Improving resolution



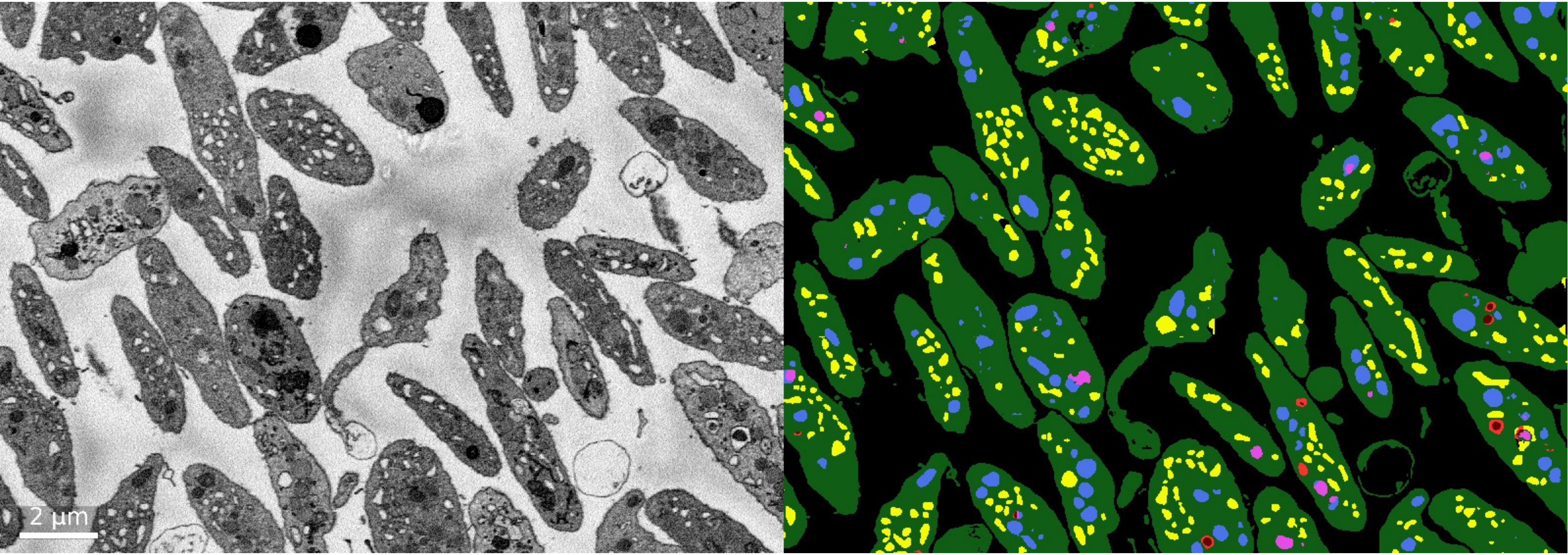
Digital Image Processing: Applications

- Texture pattern synthesis



Digital Image Processing: Applications

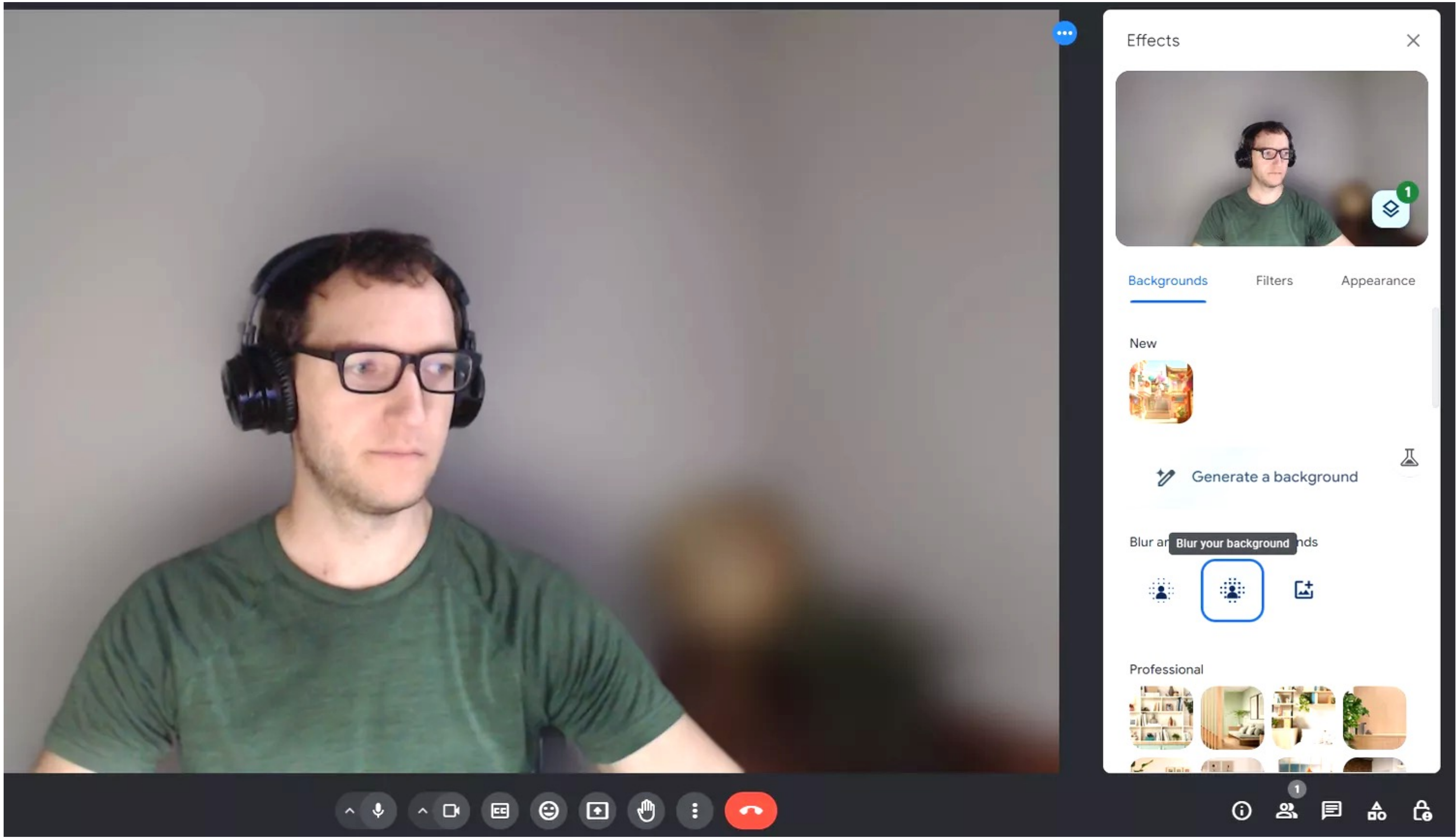
- Image segmentation



leapmanlab.github.io/dense-cell

Digital Image Processing: Applications

- Background blur



www.lifewire.com/blur-background-in-google-meet-5101054

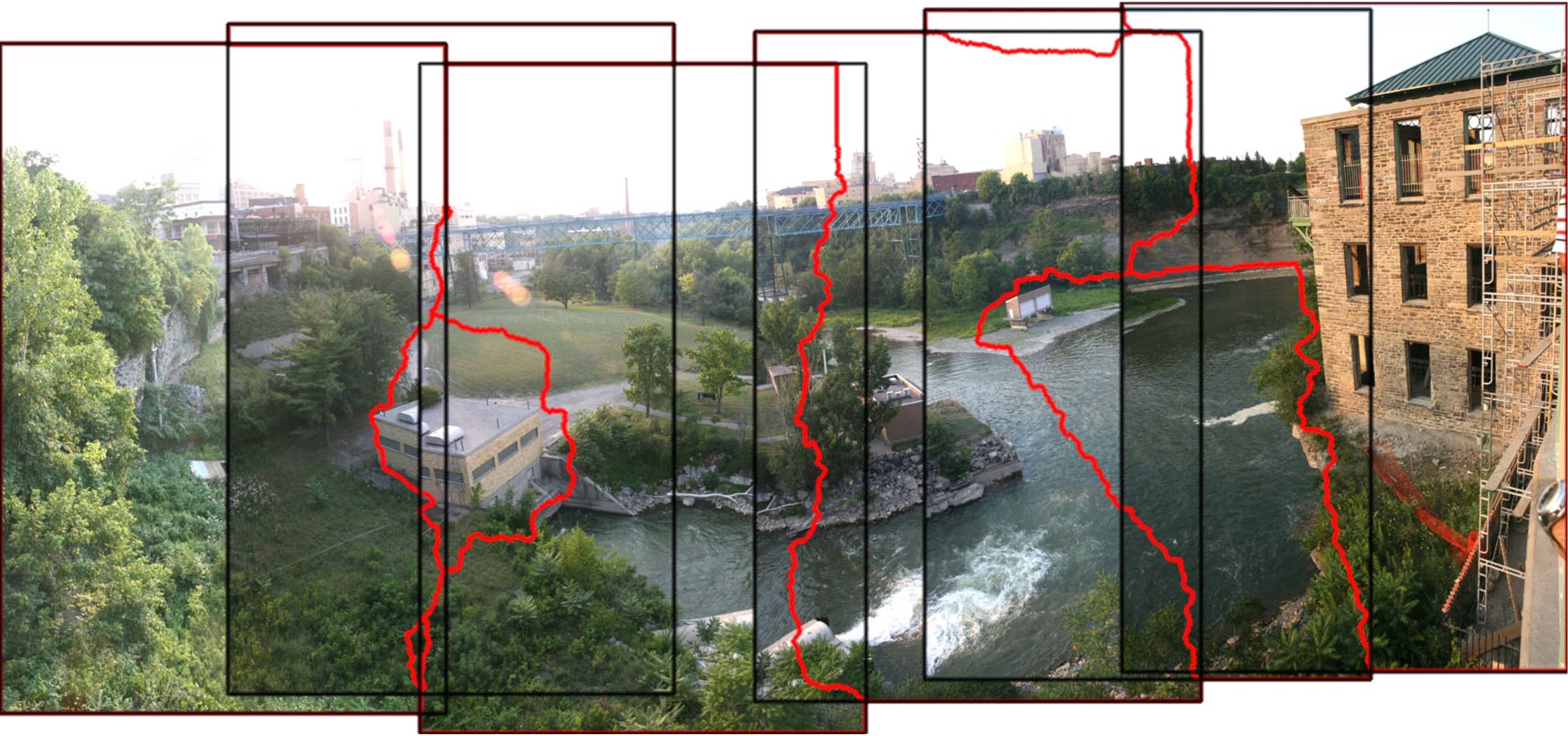
Digital Image Processing: Applications

- Virtual background

support.zoom.com/hc/en/article?id=zm_kb&sysparm_article=KB0060387

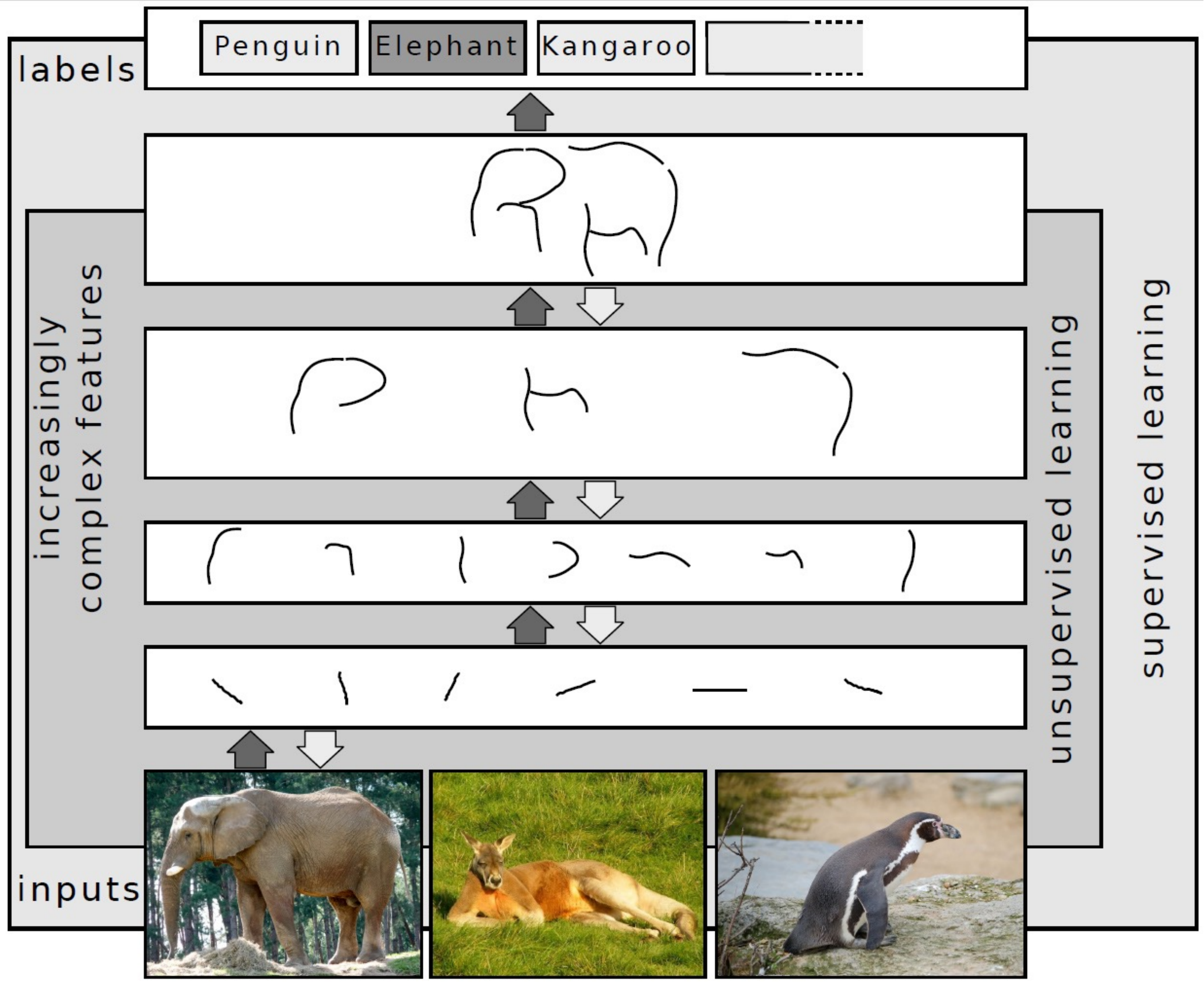
Digital Image Processing: Applications

- en.wikipedia.org/wiki/Image_stitching



Digital Image Proc

- [Machine learning](#) based methods
- [Deep learning](#) based methods
  - [Neural networks](#)
- Advantages and limitations



Digital Image Processing: Topics

1) Image acquisition
2) Basic operations, e.g., resizing, interpolation
3) Point operations and more, e.g., histogram-based processing
4) Spatial filtering, e.g., convolution, effects
5) Detecting edges and corners
6) Segmentation
7) Denoising
8) Fourier analysis
9) Principal component analysis
10) Deep neural networks